NEXUS

NETWORK MANAGEMENT SYSTEM

Complete Project Documentation

Project Details

Item	Details
Project name	NEXUS Network Management System
Technology	ASP.NET Core MVC (.NET 10), C#, Entity Framework Core, SQL Server

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Project Introduction

1. About the Project

NEXUS Network Management System is a web-based application that runs the complete business of an internet and telephone service provider working in Karachi, Lahore and Islamabad. Before such a system, the work of a service provider is scattered over paper registers, spreadsheets and phone calls: one person takes the customer's request, another checks whether the area can be served, someone installs the line, someone else prepares the bill and a different person collects the money. NEXUS brings all of these steps into one secure application, so every person sees the same up-to-date information and every action is recorded.

The system was developed by our team as the Aptech Semester 3 e-project. It is built with ASP.NET Core MVC and C#, uses Entity Framework Core with a Microsoft SQL Server database, and uses ASP.NET Core Identity for login and role-based security. The interface is designed in a dark-navy and blue theme with the Manrope font, which matches the technology and telecom identity of the NEXUS brand.

2. Problem and Objectives

A service provider must answer many questions every day: Which plan does the customer want? Is the address reachable? Which equipment was given? How much does the customer owe? Has the payment been received? NEXUS answers these questions with one connected workflow. Its main objectives are:

- to present services and plans to visitors and let customers register and order online;
- to give branch staff a controlled way to register customers and place and verify orders;
- to let the technical team record site checks, install connections and manage equipment;
- to calculate bills automatically with tax and record every payment safely;
- to give the administrator full control and live reports over staff, plans, stock and records.

3. Services and Plans

NEXUS sells three services: Broadband Internet (Home 10 Mbps, Home 25 Mbps and Business 50 Mbps plans), Dial-Up Internet (for example Dial-Up Basic with 100 free hours) and Telephone / Landline service. Each plan has a monthly charge, a one-time security deposit and an installation charge, and is managed by the administrator. Active plans are shown live on the public website.

4. The Six Parts of the System

Part	Who uses it	Main work
Public Website	Any visitor	Read about services, compare plans, contact the company, track an order or connection, register or log in.
Customer Panel	Subscribers	Place orders, view connection, plan, equipment and bills, submit payments, read notifications, send feedback.
Admin Panel	Administrator	Manage employees, customers, shops, vendors, products, stock and plans; supervise orders, connections, billing and payments.
Retail Panel	Branch staff	Register customers, create and track orders for the branch, view plans, connections, bills and payments.
Accounts Panel	Accounts staff	Generate bills, record and verify payments, send payment reminders, check customer dues.
Technical Panel	Field / network staff	Record feasibility checks, create and activate connections, assign equipment, handle repairs and equipment requests.

5. How the System Works

The heart of NEXUS is one order-to-payment workflow that passes through the panels in a fixed order:

1. A customer places an order from his panel (or retail staff place it for him). The system creates an order number such as B0000000001 with status Pending.
2. The retail branch of the customer's city verifies the customer and forwards the order to the technical team.
3. The technical team visits the site and records a feasibility report (distance, server available, Feasible or Not Feasible).
4. For a feasible order the technician creates the connection. The system generates a 16-character account number such as B001000000000001 and the equipment (modem, router, ONU) is assigned from stock.
5. The accounts team generates the monthly bill: plan charge plus equipment charges minus discount, plus 12.24% service tax and any previous balance.
6. The customer pays at the branch or submits an online payment from his panel; the accounts team verifies it and only then are the bill's paid and due amounts updated.
7. At every step the customer can see progress in his panel, and the admin can supervise everything from the dashboard.

Order-to-payment workflow of NEXUS

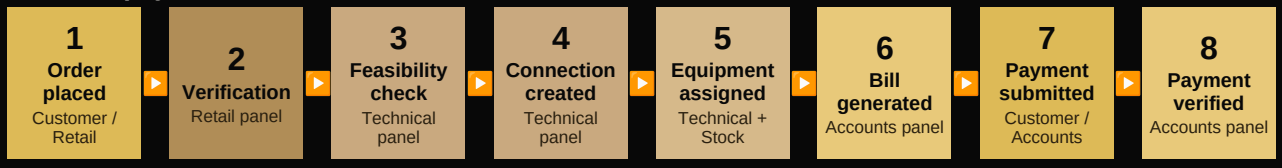


Figure: Complete order-to-payment workflow; colours show the panel that performs each step

6. Important Business Rules

- **Order number:** D, B or T (Dial-Up, Broadband, Telephone) followed by a 10-digit sequence, for example B0000000001. It is always generated by the system.
- **Account number:** type letter + 3-digit city code + 12-digit serial, for example B001000000000001 (16 characters, unique).
- **Bill:** plan charge + equipment and replacement charges - discount, plus 12.24% service tax and any previous unpaid balance; the due date is 14 days after the billing period ends.
- **Payment:** an online payment never changes the bill by itself; it is created as PendingVerification and updates the bill only after the accounts team verifies it.
- **Stock:** products ordered with a connection are reserved immediately, released if the order is cancelled, and available stock = on hand - reserved.
- **Order status:** an order moves only through allowed steps (Pending, VerifiedByRetail, Feasible, ConnectionProvided and so on) and every change is stored in the status history.

7. Technology Used

Layer	Technology
Application	ASP.NET Core MVC with Areas (.NET 10), C#
User interface	Razor views, HTML, CSS and JavaScript with a custom NEXUS theme (Manrope font)
Database	Microsoft SQL Server with Entity Framework Core (Code First, migrations)
Security	ASP.NET Core Identity, five roles, account lockout, anti-forgery tokens, login audit log
Email	Resend email API (used for replies to contact messages and order notices)

8. Key Features and Security

- Five roles (Admin, Retail, Technical, Accounts, Customer); each user can open only his own panel, and customers see only their own records.
- Automatic order and account numbers, automatic bill calculation and stock reservation for ordered equipment.
- Payments submitted online stay PendingVerification until the accounts team approves them.

- Complete history: order status history, plan history, stock movements and login attempts are all stored.
- Passwords need 8 or more characters with upper-case, lower-case and a digit; five wrong attempts lock the account for 15 minutes.
- Records that are already used are deactivated instead of deleted, so reports and history stay correct.

9. Modules (Menu Pages) of Each Panel

Panel	Pages / modules
Public Website	Home, Services, Plans, Plan Details, About, Contact, Track Order, Connection Status, Login, Register
Customer Panel	Dashboard, Profile, My Orders, My Connection, Current Plan, My Equipment, My Bills, Payment History, Notifications, Feedback
Admin Panel	Dashboard, Employees, Customers, Retail Shops, Vendors, Products, Stock, Plans, Orders, Connections, Billing, Payments, Messages, Feedback, Advanced Search
Retail Panel	Dashboard, Customers, Orders (with Track Order), Plans, Connections, Bills, Payments
Accounts Panel	Dashboard, Bills (Generate Bill), Payments, Outstanding Dues, Customers, Connections
Technical Panel	Dashboard, Orders, Feasibility, Connections, Equipment, Equipment Requests

10. Conclusion

NEXUS turns the daily work of a telecom service provider into one clear, secure and traceable process. A customer’s request can be followed from the first order to the last payment, each department works in its own panel with only the tools it needs, and management gets live figures instead of manual reports. The project demonstrates the complete use of ASP.NET Core MVC, Entity Framework Core, SQL Server and role-based security in a real business scenario.

11. How to Read This Document

After this introduction, every screenshot of the application is shown with a short explanation of what that page does. The order is: Public Website, Customer Panel, Admin Panel, Retail Panel, Accounts Panel and Technical Panel. The document ends with the database design: an image of all tables, the list of table names and the complete Entity Relationship Diagram (ERD).

Public Website

The public website is the face of NEXUS. Any visitor can read about the company, compare plans, check an order or connection, and send a message without logging in. Plans shown here are read live from the database, so whenever the admin changes a plan it updates on the website automatically.

Home Page - Hero Banner

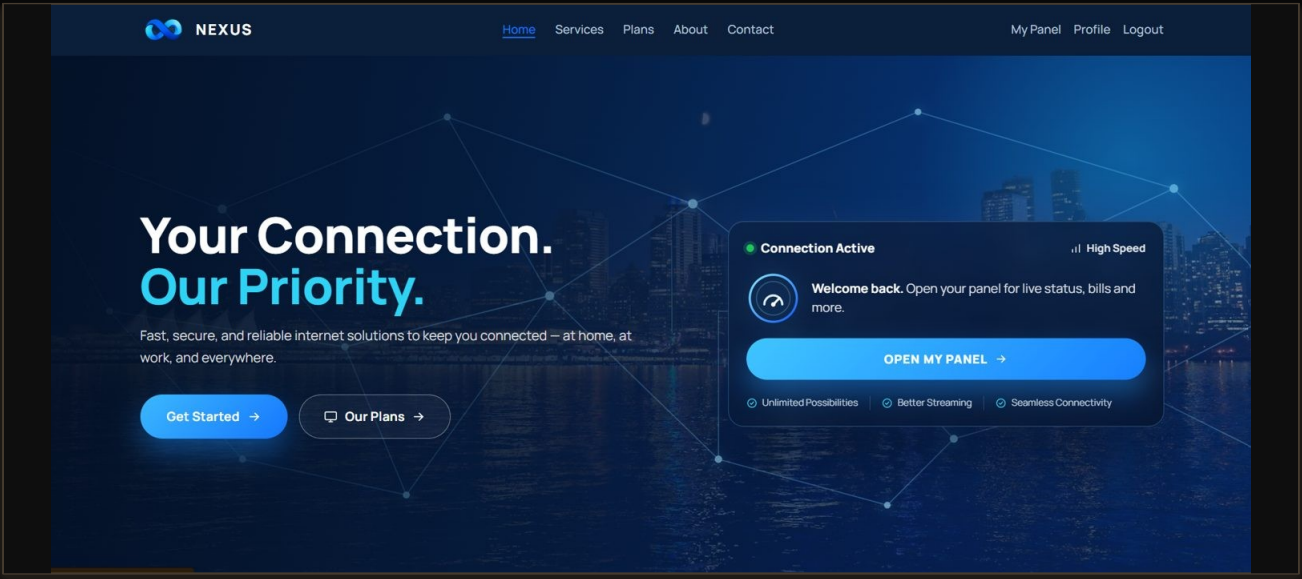


Figure 1: Home Page - Hero Banner

Purpose: This is the first screen of the website. The headline “Your Connection. Our Priority.” explains the brand promise, and the two buttons “Get Started” and “Our Plans” push the visitor to register or browse plans. The glass card on the right shows a quick “Open My Panel” button which sends a signed-in user straight to his own panel (customer, retail, technical, accounts or admin). The navigation bar at the top links to Home, Services, Plans, About and Contact.

Home Page - Services Overview

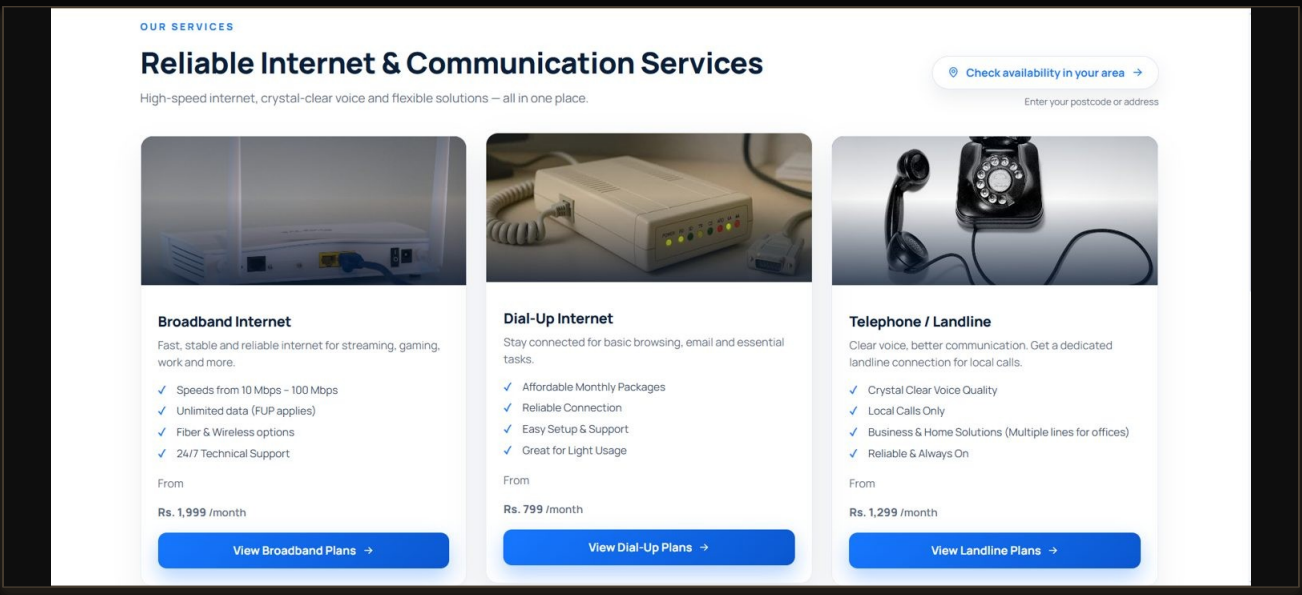


Figure 2: Home Page - Services Overview

Purpose: This section introduces the three services sold by NEXUS: Broadband Internet, Dial-Up Internet and Telephone / Landline. Each card lists the main features (speed range, unlimited data with fair-usage policy, 24/7 support, local calls) and has a button that opens the plans of that service. The “Check availability in your area” link lets a visitor verify the service before ordering.

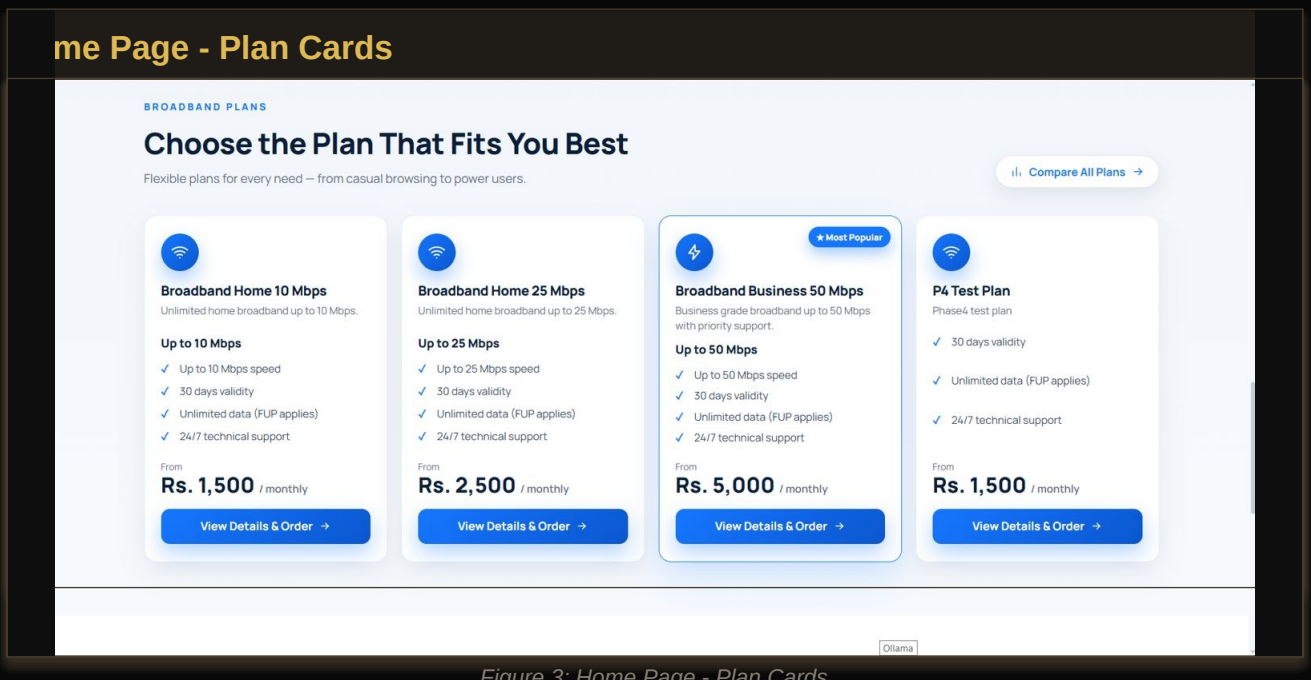


Figure 3: Home Page - Plan Cards

Purpose: Here the active Broadband plans are displayed as cards: Broadband Home 10 Mbps (Rs. 1,500), Home 25 Mbps (Rs. 2,500), Business 50 Mbps (Rs. 5,000, marked Most Popular) and the test plan. Every card shows speed, validity, data policy, support and the monthly price. “View Details & Order” opens the full plan page, and “Compare All Plans” opens the Plans page.

Services Page

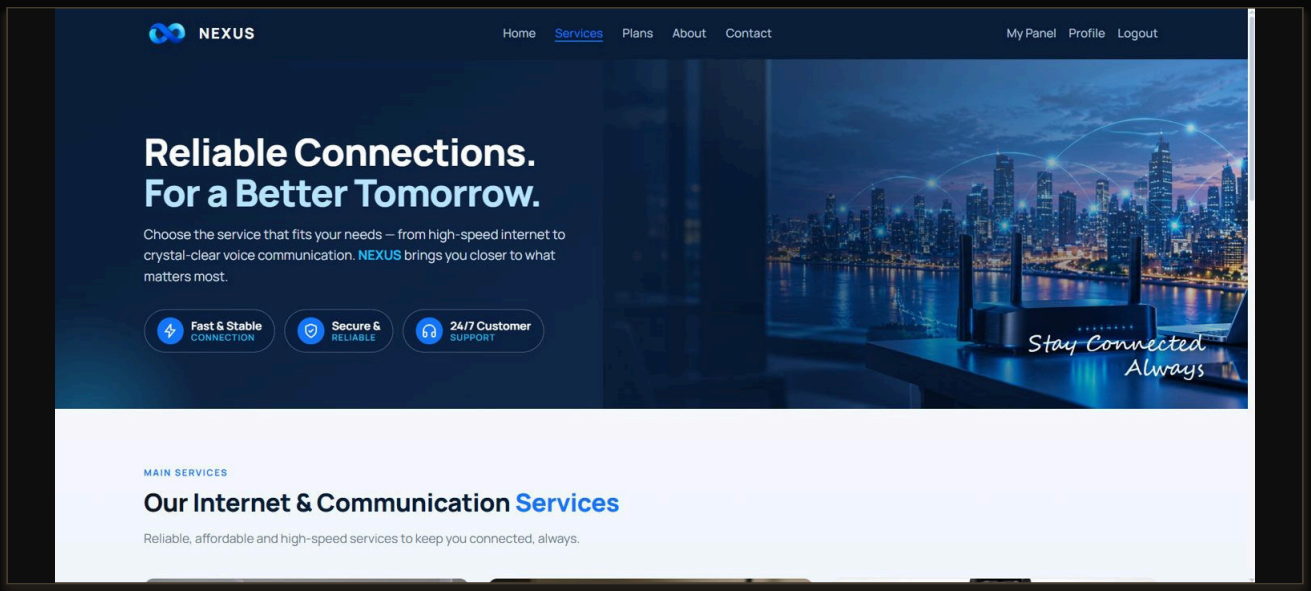


Figure 4: Services Page

Purpose: The Services page explains what NEXUS offers in more detail. The hero banner “Reliable Connections. For a Better Tomorrow.” carries three trust badges (Fast & Stable Connection, Secure & Reliable, 24/7 Customer Support), and the section below describes each internet and communication service with pictures.

Plans Page - Banner and Dial-Up Plans

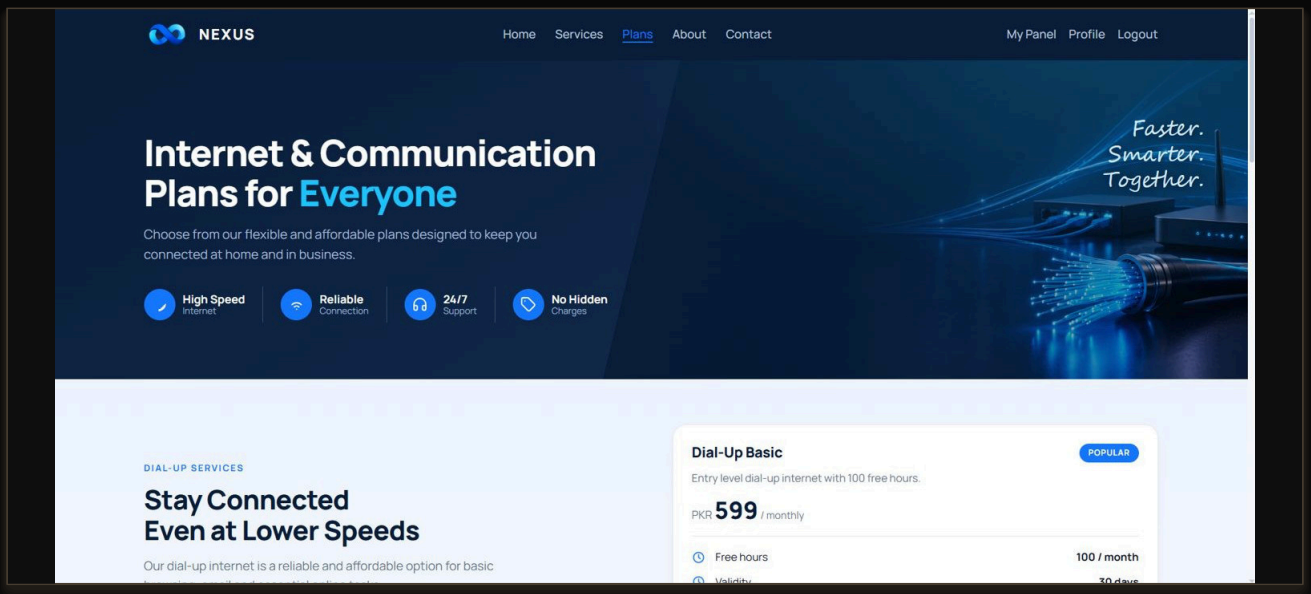


Figure 5: Plans Page - Banner and Dial-Up Plans

Purpose: The Plans page is the complete plan catalogue. The banner promises high speed, reliability, 24/7 support and no hidden charges. The first group shown is Dial-Up Services, where the Dial-Up Basic plan (PKR 599 per month, 100 free hours, 30 days validity) is displayed with a Popular tag for customers who need a low-cost connection.

Plans Page - Broadband Plans

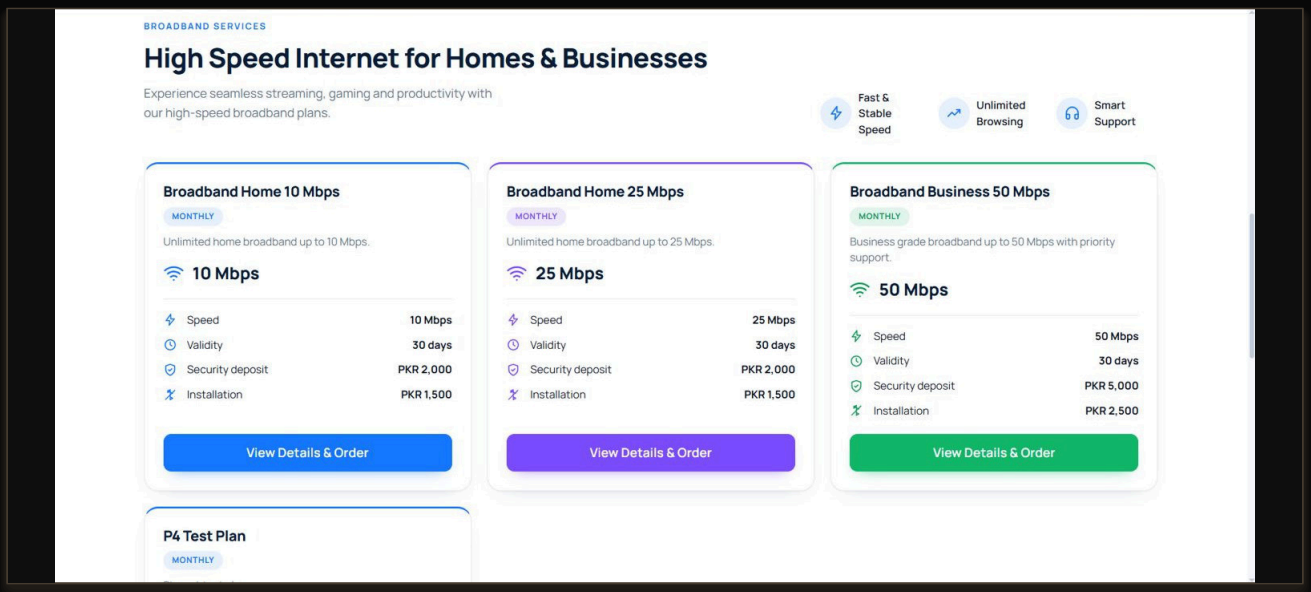


Figure 6: Plans Page - Broadband Plans

Purpose: This part of the Plans page lists every Broadband plan in colour-coded cards. Each card compares speed, validity, security deposit and installation charge (for example Home 10 Mbps: PKR 2,000 deposit and PKR 1,500 installation), so the visitor knows all one-time and monthly costs before ordering. The “View Details & Order” button opens the Plan Details page.

Plan Details Page

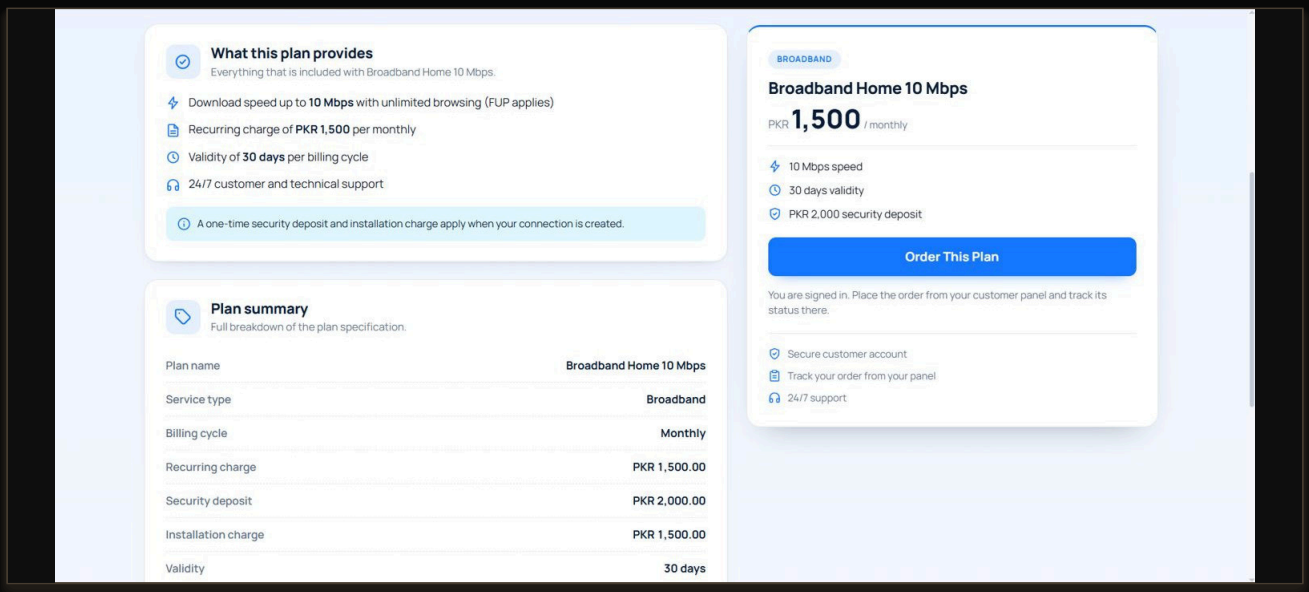


Figure 7: Plan Details Page

Purpose: This page shows the complete specification of one selected plan. “What this plan provides” lists speed, monthly charge, validity and support, and the Plan summary table lists service type, billing cycle, recurring charge, security deposit and installation charge. The “Order This Plan” button opens the order form with this plan already selected; a visitor who is not signed in is asked to log in or register first.

About Page

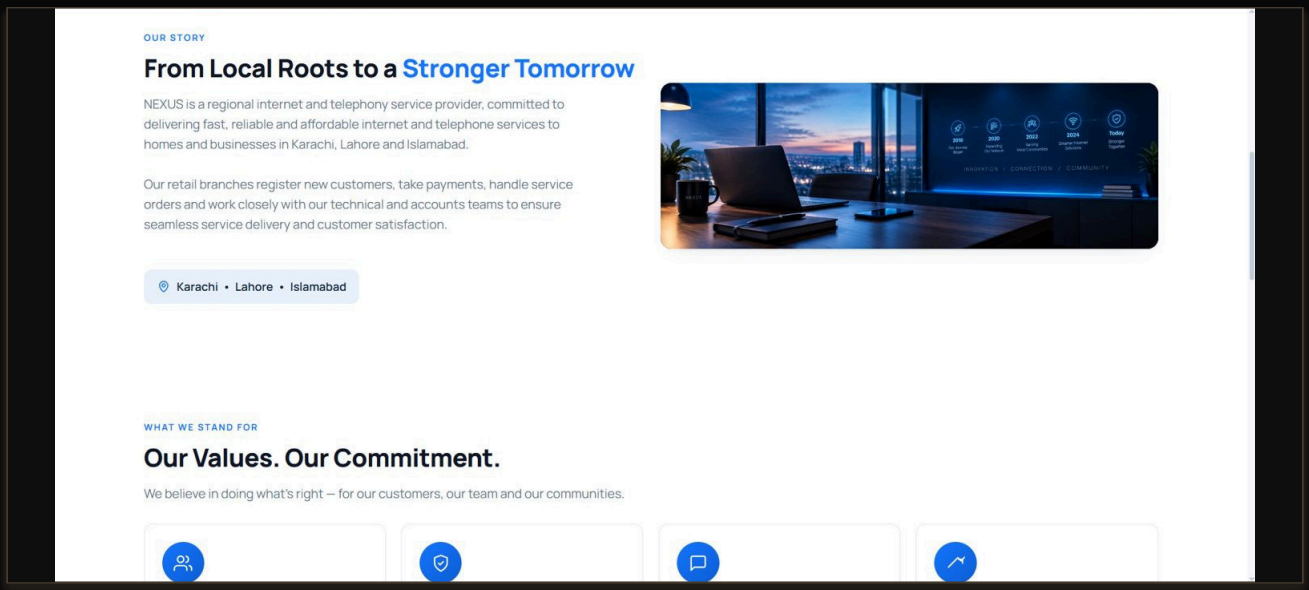


Figure 8: About Page

Purpose: The About page tells the story of NEXUS as a regional internet and telephony provider serving Karachi, Lahore and Islamabad. It explains how retail branches, the technical team and the accounts team work together, and lists the company values (People First, Dependable Service, Clear Communication and Continuous Improvement) to build trust with visitors.

Contact Page

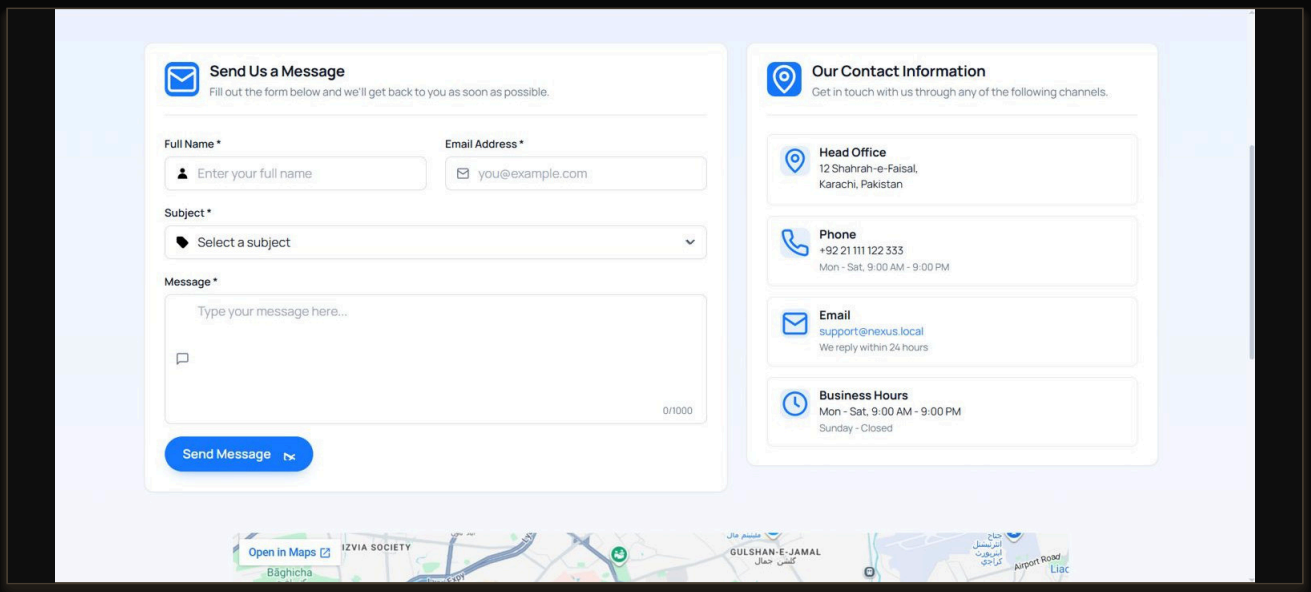


Figure 9: Contact Page

Purpose: The Contact page lets a visitor send a message using Full Name, Email, Subject and Message (up to 1000 characters). The message is stored in the database and appears in the Admin Messages inbox, where the admin can reply by email. The right-hand card shows the head office address, phone number, support email and business hours, and a map below helps visitors find the office.

Customer Panel

The Customer Panel is the self-service area for subscribers. After login a customer can place a service order, follow it until the connection is installed, see his connection, plan and equipment, read bills, submit payments, receive messages from the accounts team and send feedback. Every query is filtered by the logged-in customer, so one customer can never see another customer's data.

Customer Dashboard

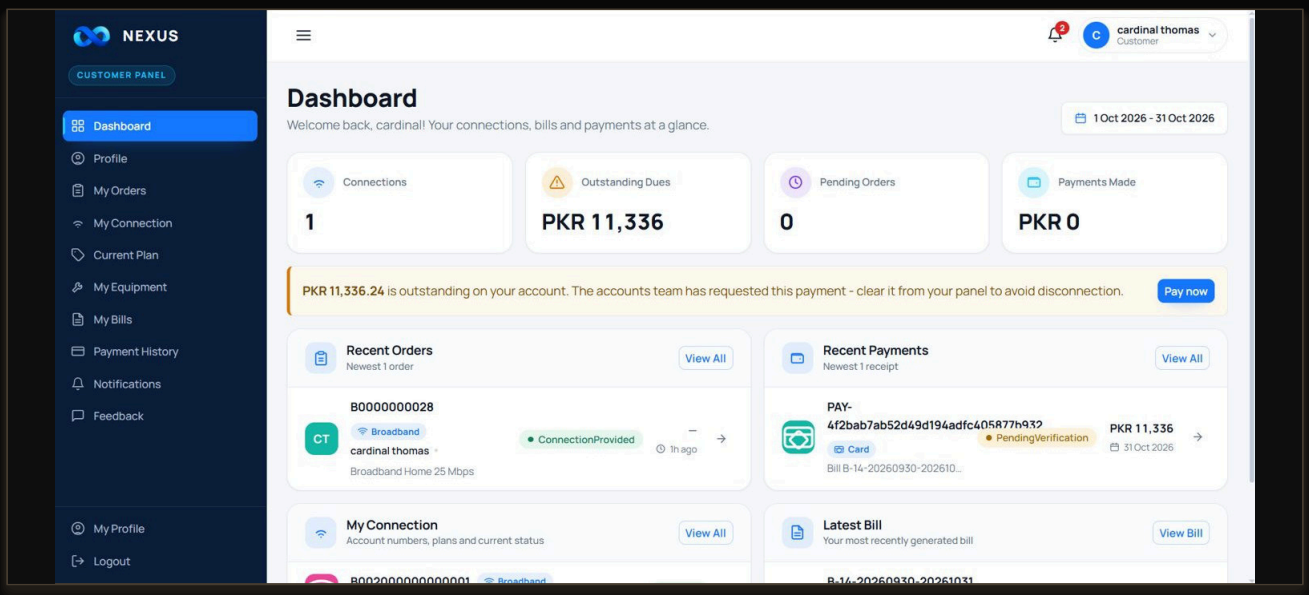


Figure 10: Customer Dashboard

Purpose: The dashboard gives a one-screen summary of the account: number of connections, outstanding dues, pending orders and payments made. A yellow alert shows the amount due with a "Pay now" button. The cards below show recent orders, the latest payment, the connection and the most recent bill. Payments waiting for accounts approval are not counted as paid, which is why Payments Made shows PKR 0.

My Profile

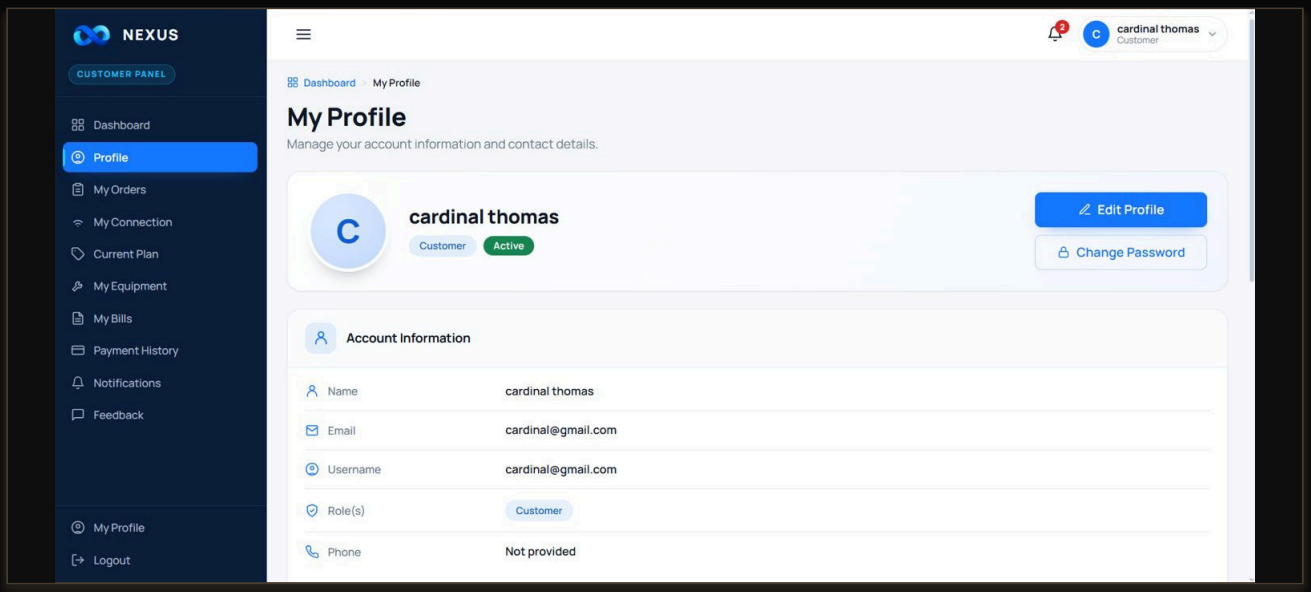


Figure 11: My Profile

Purpose: The profile page shows the customer's name, email, username, role and phone number with a green Active badge. "Edit Profile" updates contact and address details and "Change Password" changes the login password after confirming the current one. The email is read-only because it is the login name.

My Orders

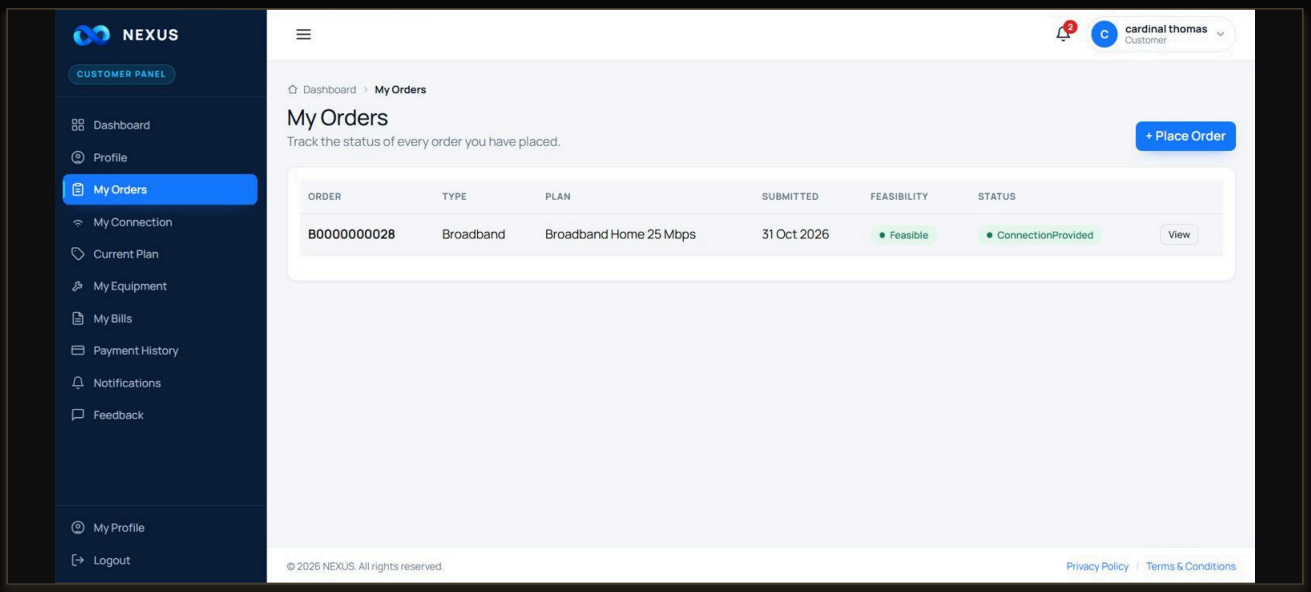


Figure 12: My Orders

Purpose: This page lists every order placed by the customer with order number, type, plan, submission date, feasibility result and status (for example ConnectionProvided). The View button opens the order details with products, feasibility result and the connection created from it. "+ Place Order" starts a new request.

Place a Service Order

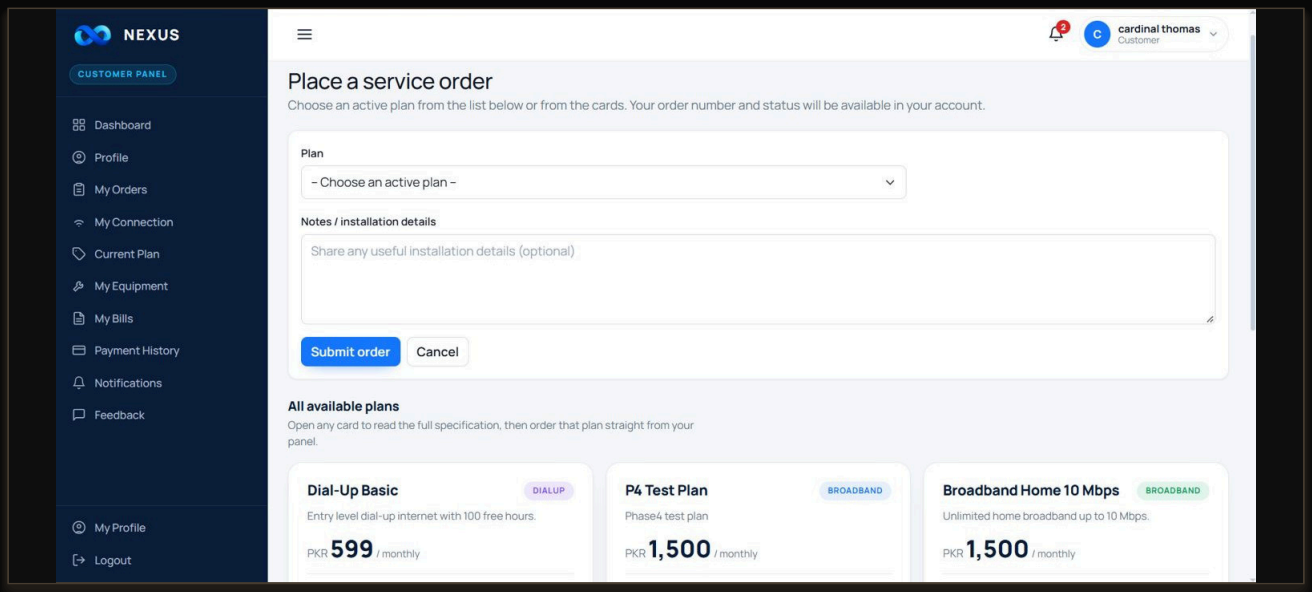


Figure 13: Place a Service Order

Purpose: The customer chooses an active plan, writes optional installation notes and presses Submit order. The system generates the order number automatically, sets the status to Pending and routes the order to the retail branch of the customer's city. All available plans are also shown as cards below the form for easy comparison.

My Connection

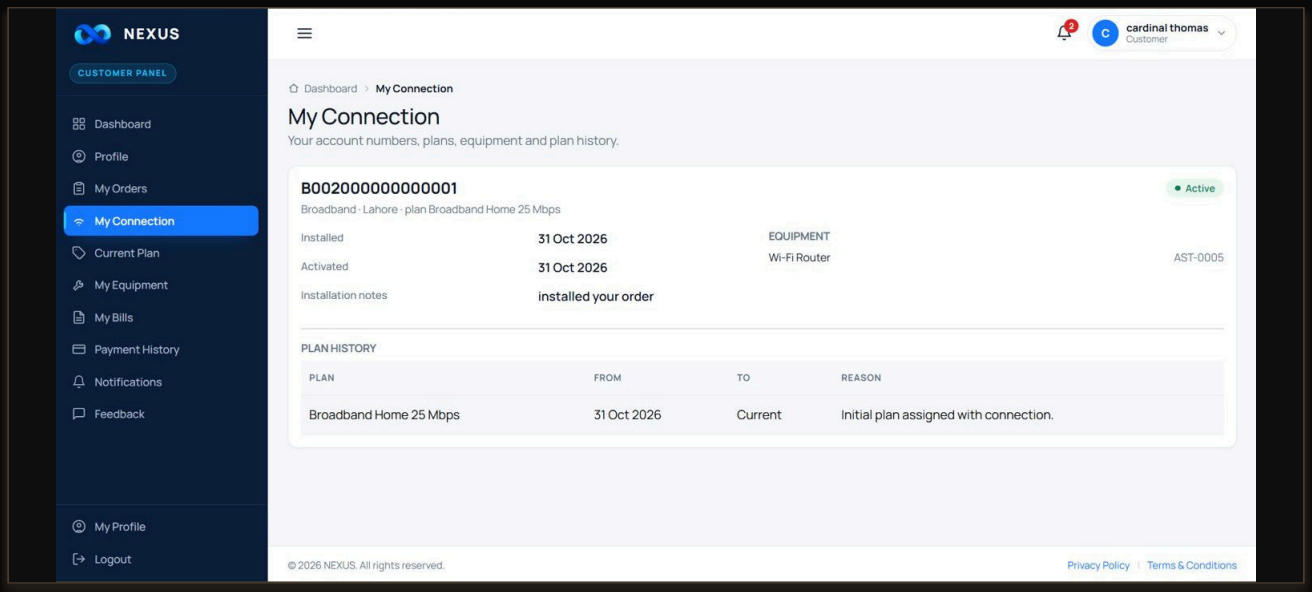


Figure 14: My Connection

Purpose: This page shows the connection provided to the customer: the 16-character account number, service type, city, plan, installation and activation dates, installation notes written by the technician and the equipment assigned (for example a Wi-Fi router). The Plan History table records every plan the connection has had and why it changed.

Current Plan

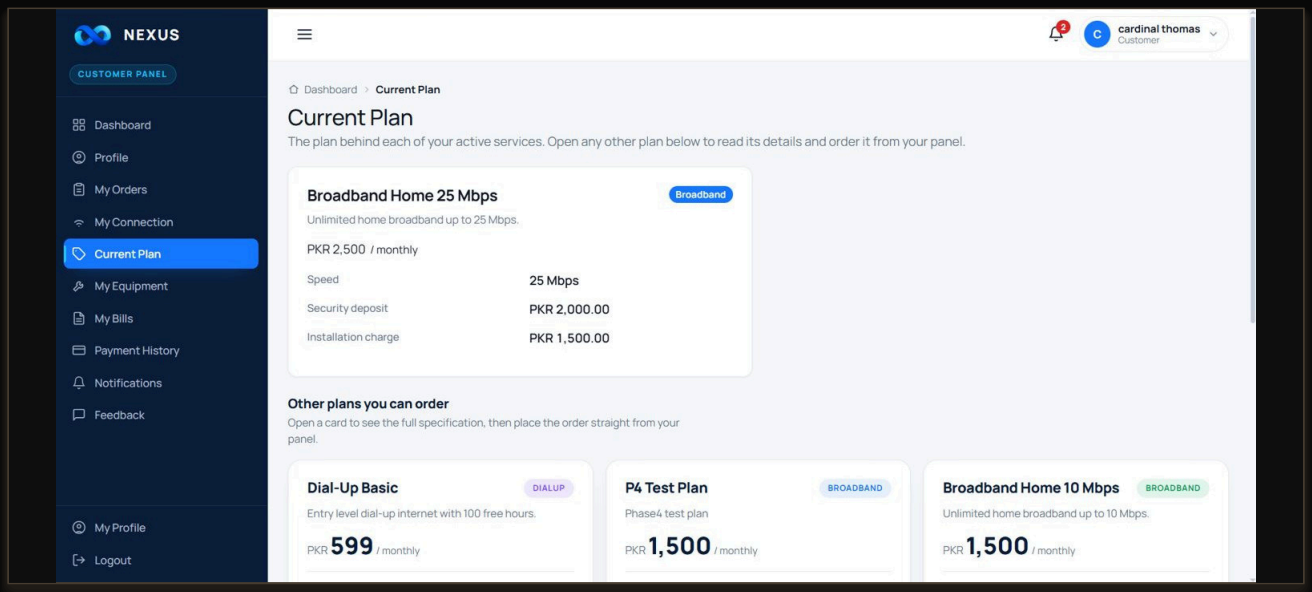


Figure 15: Current Plan

Purpose: This page explains the plan behind the customer's active service (speed, monthly price, security deposit and installation charge) and lists the other plans that the customer can order from his own panel.

My Equipment

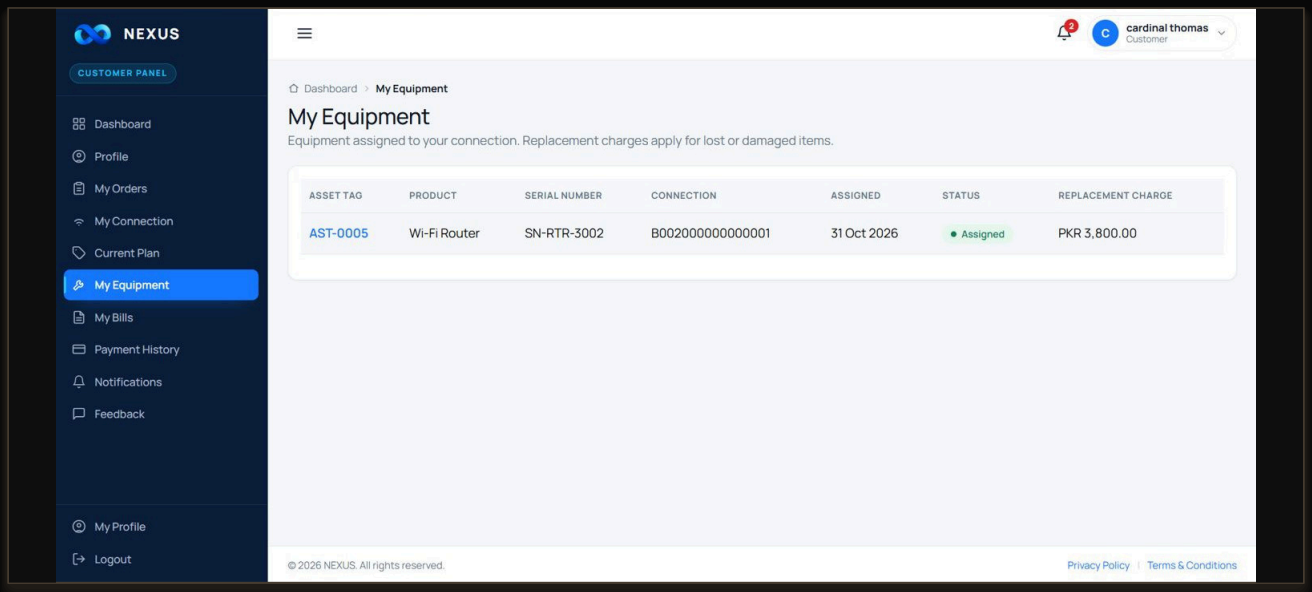


Figure 16: My Equipment

Purpose: The equipment page lists the devices installed at the customer's connection with asset tag, product, serial number, assigned date, status and the replacement charge that applies if the item is lost or damaged. From the equipment details page the customer can send a cancellation or return request, which the technical team reviews.

My Bills

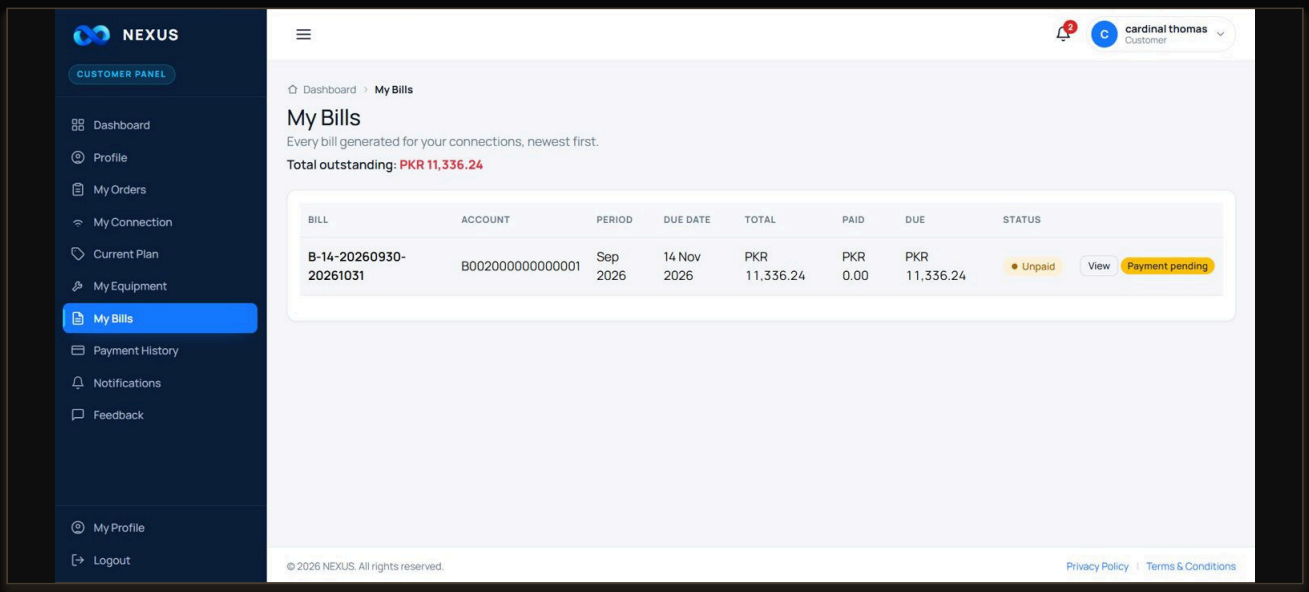


Figure 17: My Bills

Purpose: All bills generated for the customer are listed newest first with bill number, account, period, due date, total, paid, due and status. The total outstanding amount is highlighted in red. A “Payment pending” tag appears when the customer has already submitted a payment that accounts has not yet confirmed.

Bill Details

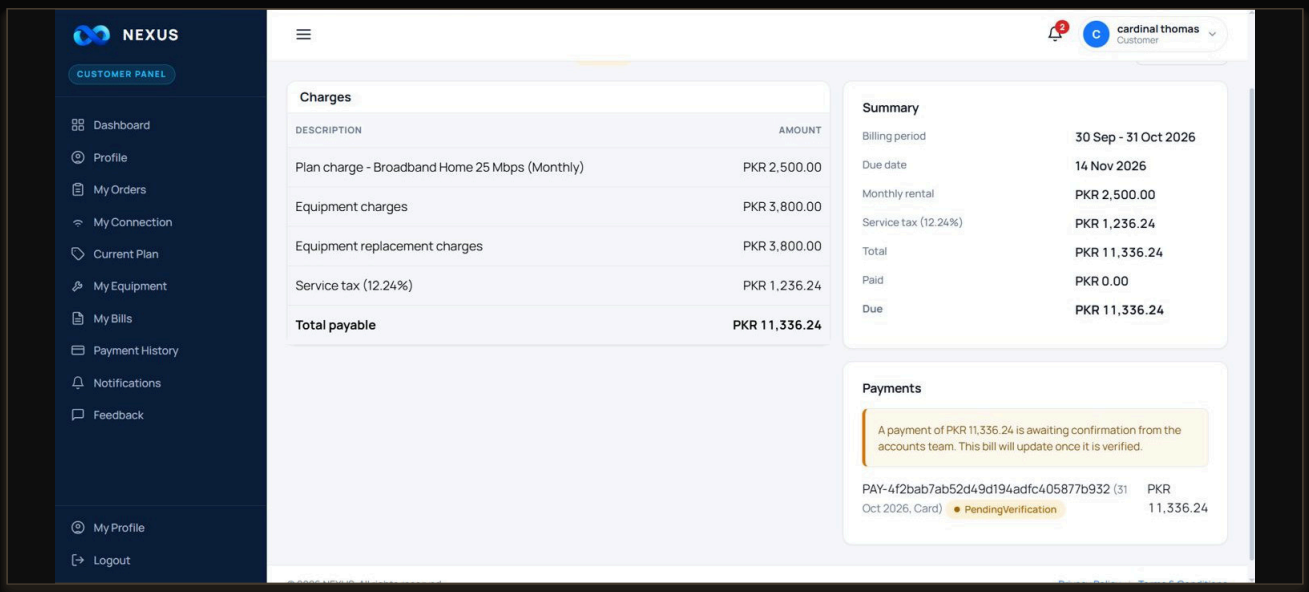


Figure 18: Bill Details

Purpose: The bill detail page itemises the charges: plan charge, equipment charges, replacement charges and 12.24% service tax, ending with the total payable. The summary card shows billing period, due date, paid and due amounts, and the Payments card shows any submitted payment and its verification status.

Payment History

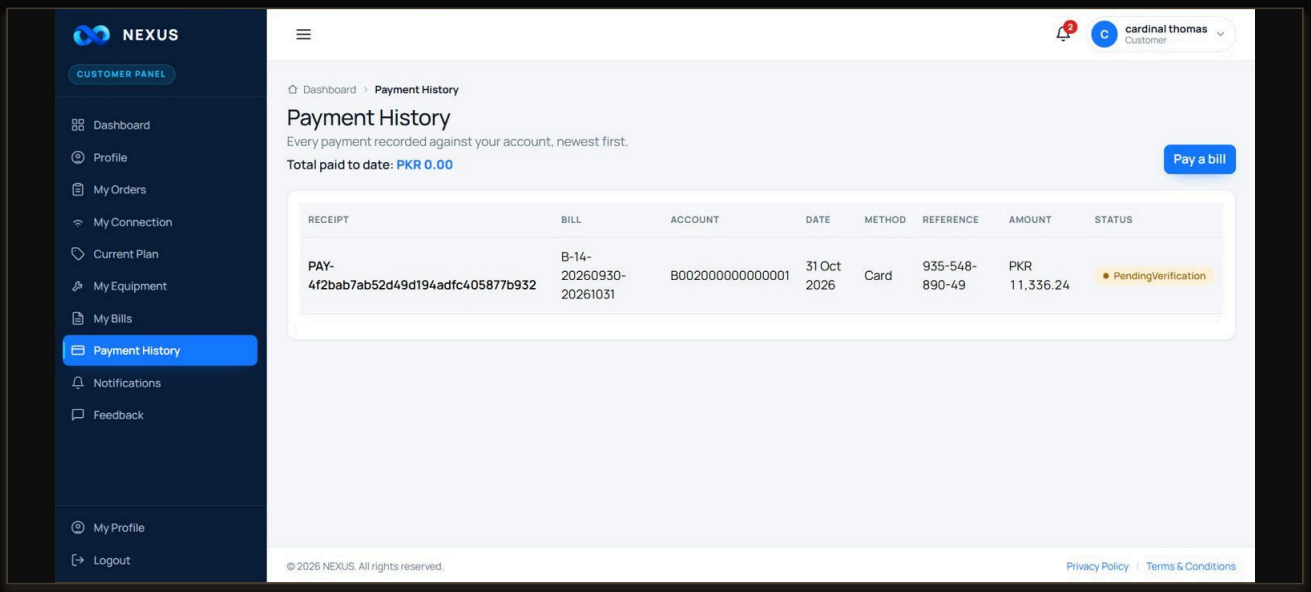


Figure 19: Payment History

Purpose: This page lists every payment the customer has submitted or made, with receipt number, bill, account, date, method, reference and status. “Total paid to date” counts only confirmed payments; a payment marked PendingVerification is waiting for the accounts team. The “Pay a bill” button opens the payment form.

Pay a Bill

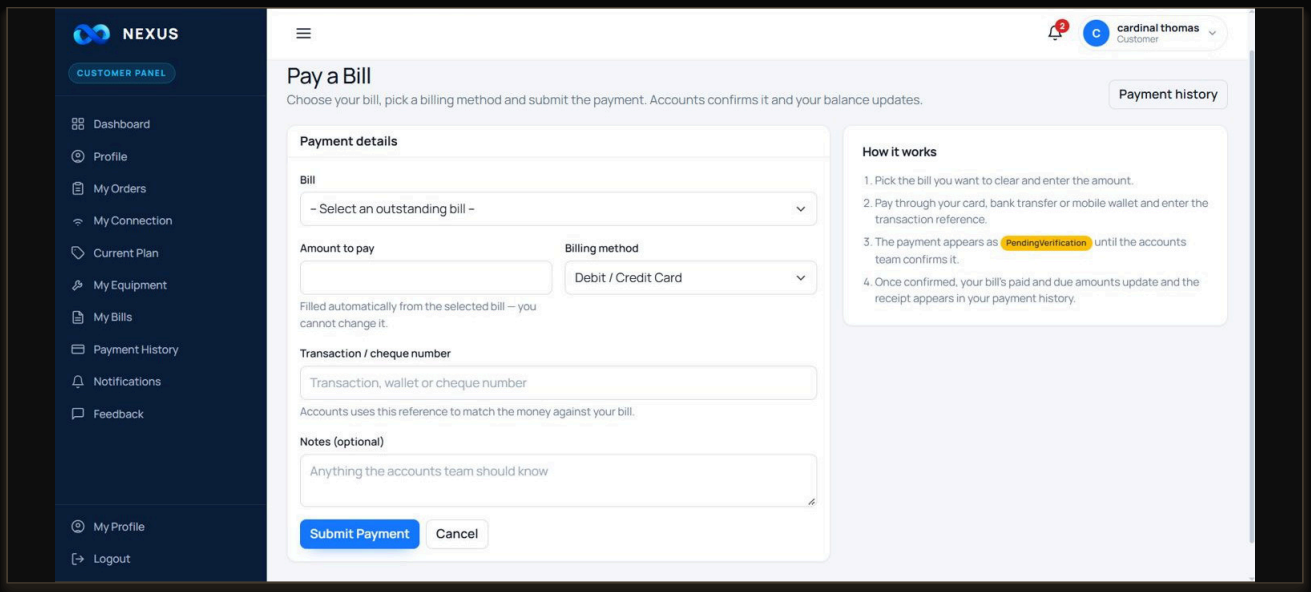


Figure 20: Pay a Bill

Purpose: The customer selects an outstanding bill, the amount is filled automatically and cannot be edited, then chooses the billing method (card, bank transfer, mobile wallet or cheque) and enters the transaction reference. After submission the payment stays PendingVerification until the accounts team confirms it; only then do the bill’s paid and due amounts change.

Notifications

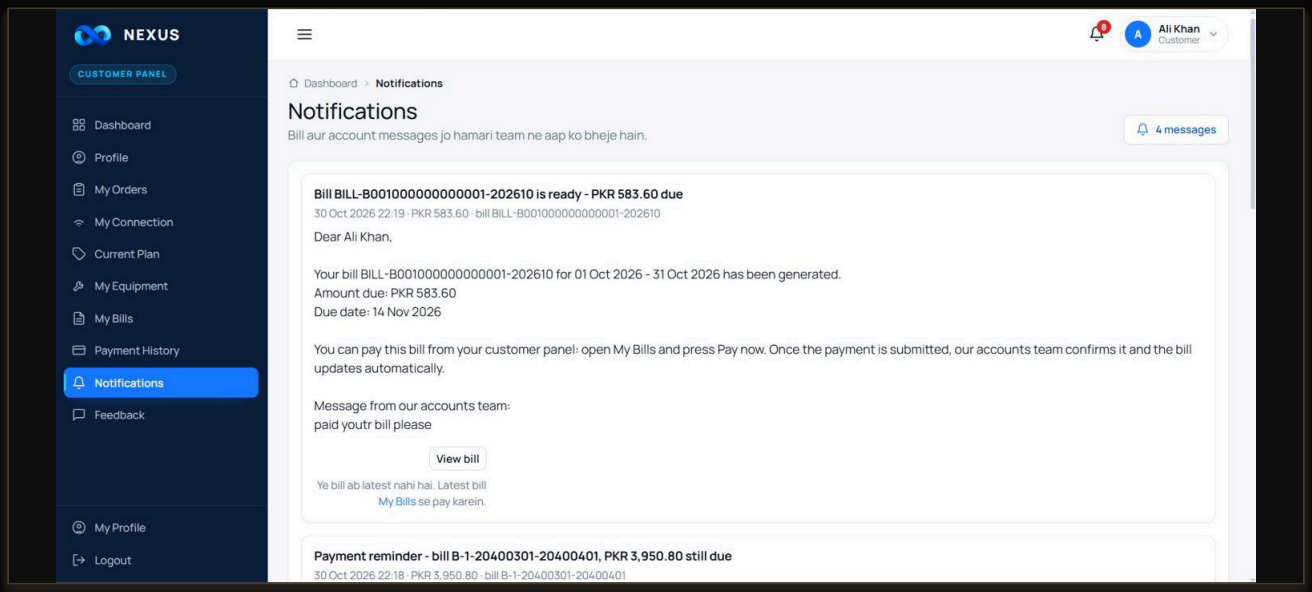


Figure 21: Notifications

Purpose: Messages sent by the accounts team appear here, such as “bill is ready” notices and payment reminders, with amount, due date and an optional personal note. A “View bill” button opens the bill, and the red counter on the bell icon shows unread messages. Messages are marked as read when the page is opened.

Feedback

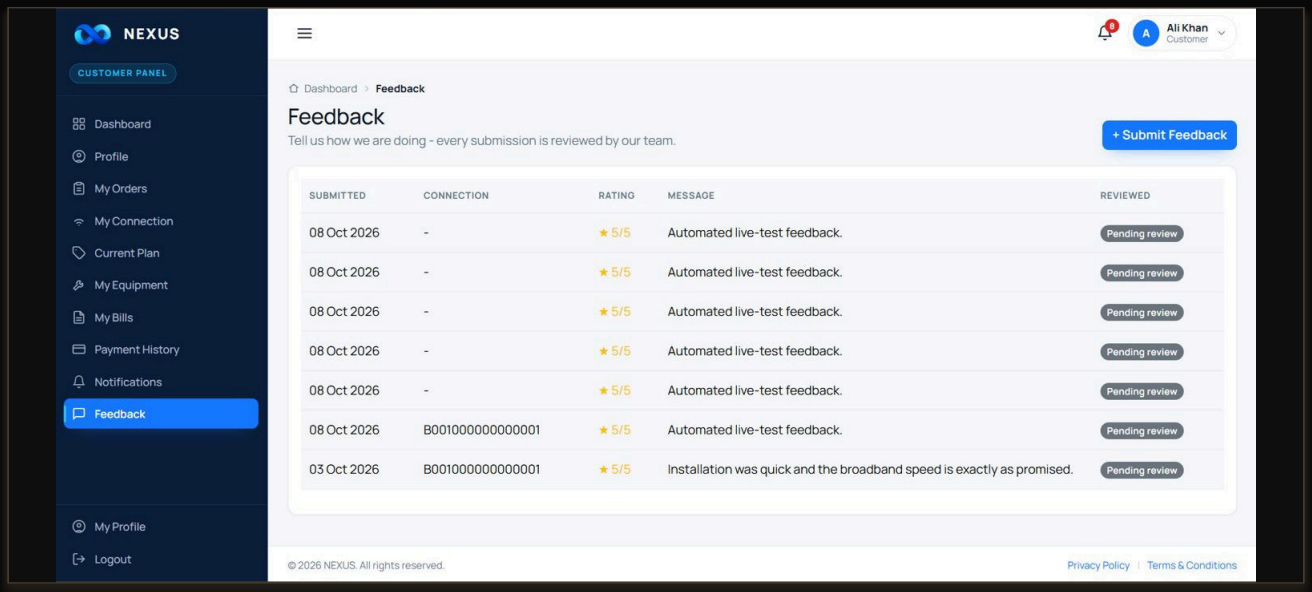


Figure 22: Feedback

Purpose: Customers can rate the service from 1 to 5 stars and write a comment, optionally linked to one of their connections. The list shows date, connection, rating, message and review status; “Pending review” changes to Reviewed when the admin reads the entry.

Admin Panel

The Admin Panel is the control centre of the company. The administrator manages employees, customers, retail shops, vendors, products, stock and plans, supervises every order, connection, bill and payment, reads customer messages and feedback, and can search the whole system from the top search bar. Records that are already used elsewhere are deactivated instead of deleted so history is never lost.

Admin Dashboard

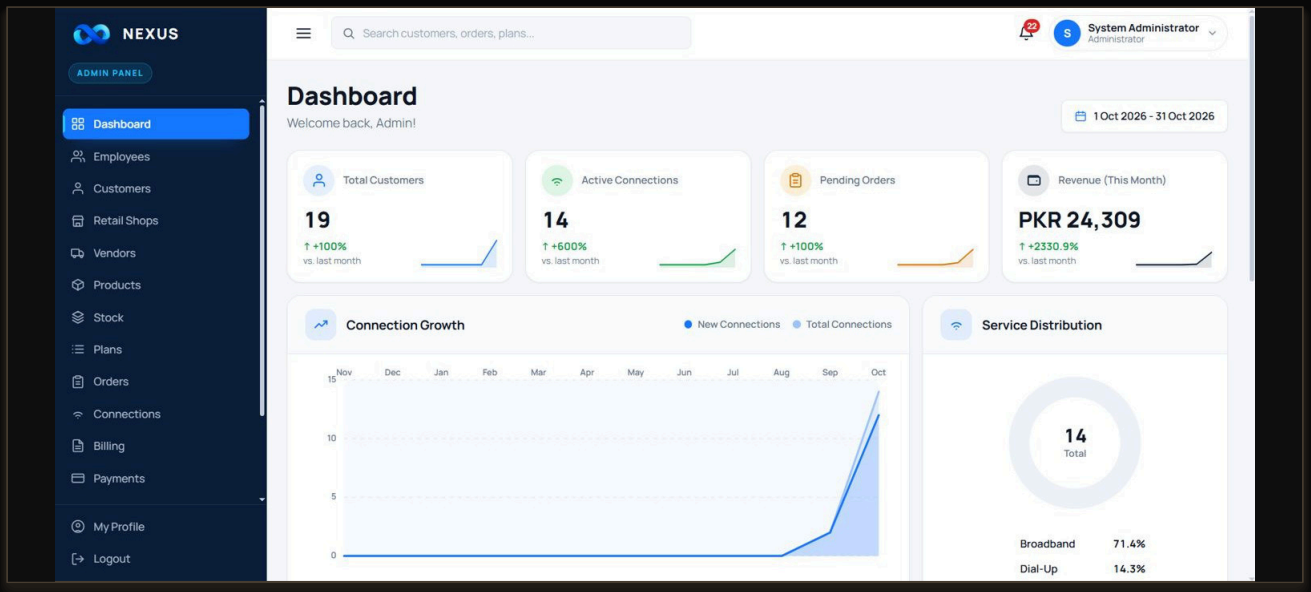


Figure 23: Admin Dashboard

Purpose: The dashboard shows live management figures: total customers, active connections, pending orders and revenue for the month, each with a trend against last month. The Connection Growth chart plots new and total connections for the last twelve months, and the Service Distribution donut shows the share of Broadband, Dial-Up and Telephone connections.

Employees

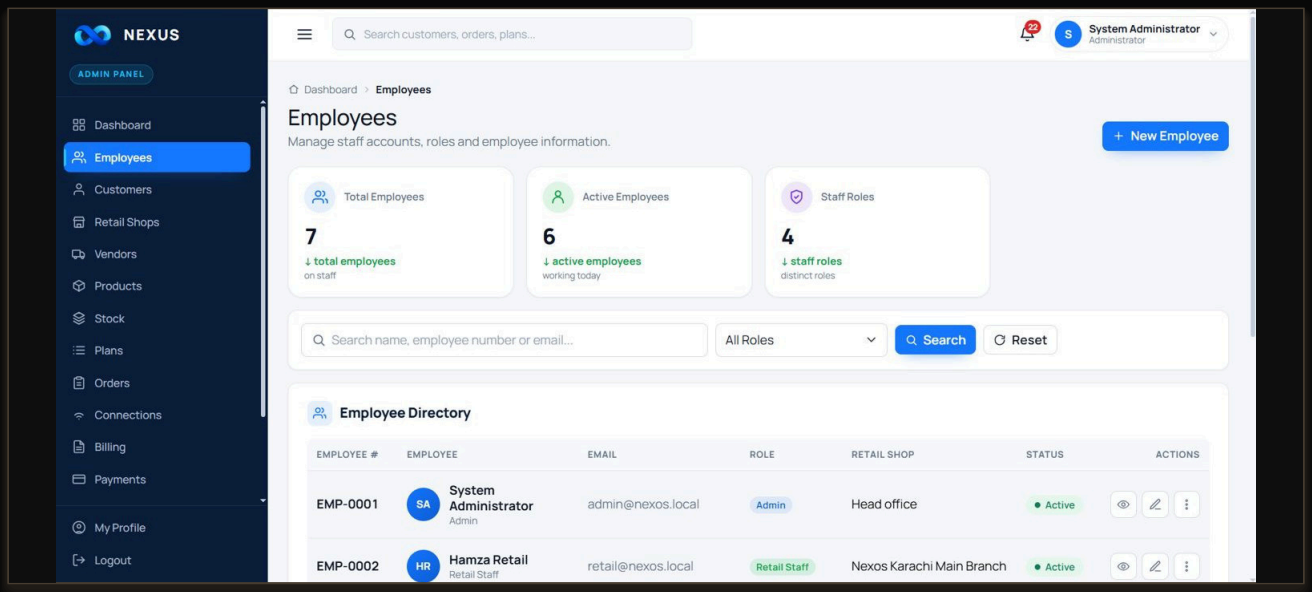


Figure 24: Employees

Purpose: This module manages staff. The cards show total employees, active employees and staff roles, and the directory lists employee number, name, email, role (Admin, Retail, Technical or Accounts), retail shop and status. Creating an employee also creates the login account with the correct role in one step; deleting only deactivates the employee.

Customers

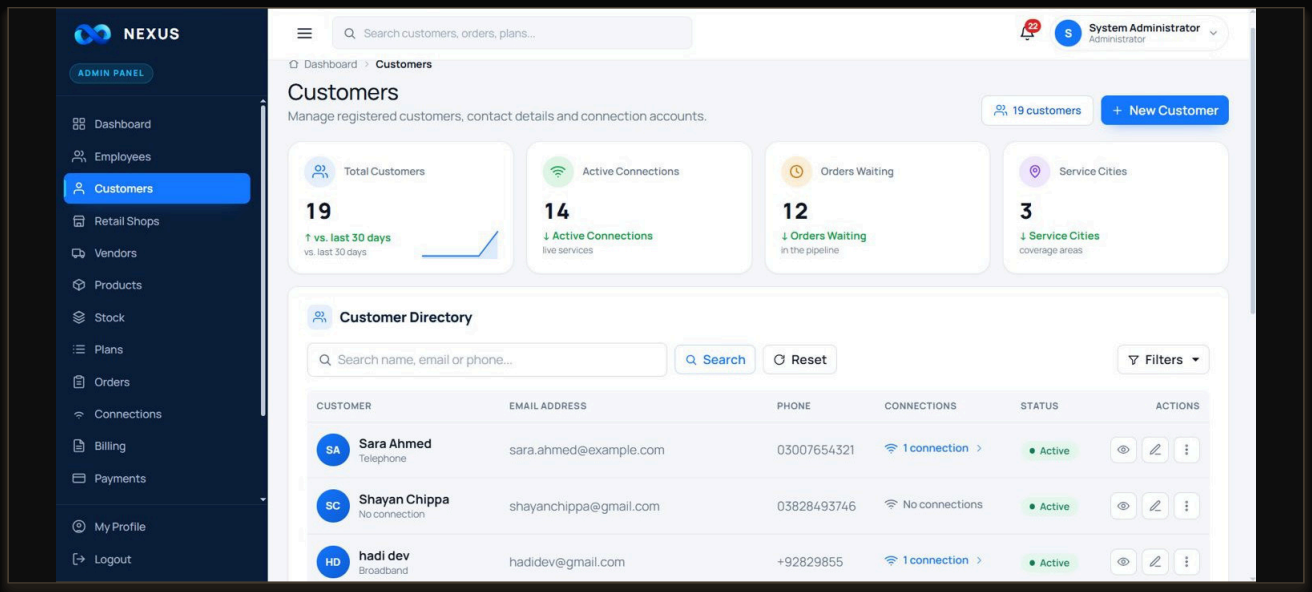


Figure 25: Customers

Purpose: The customer directory shows every registered customer with email, phone, number of connections and status, plus summary cards for total customers, active connections, orders waiting and service cities. The admin can search, filter, add, view, edit or deactivate customers.

Retail Shops

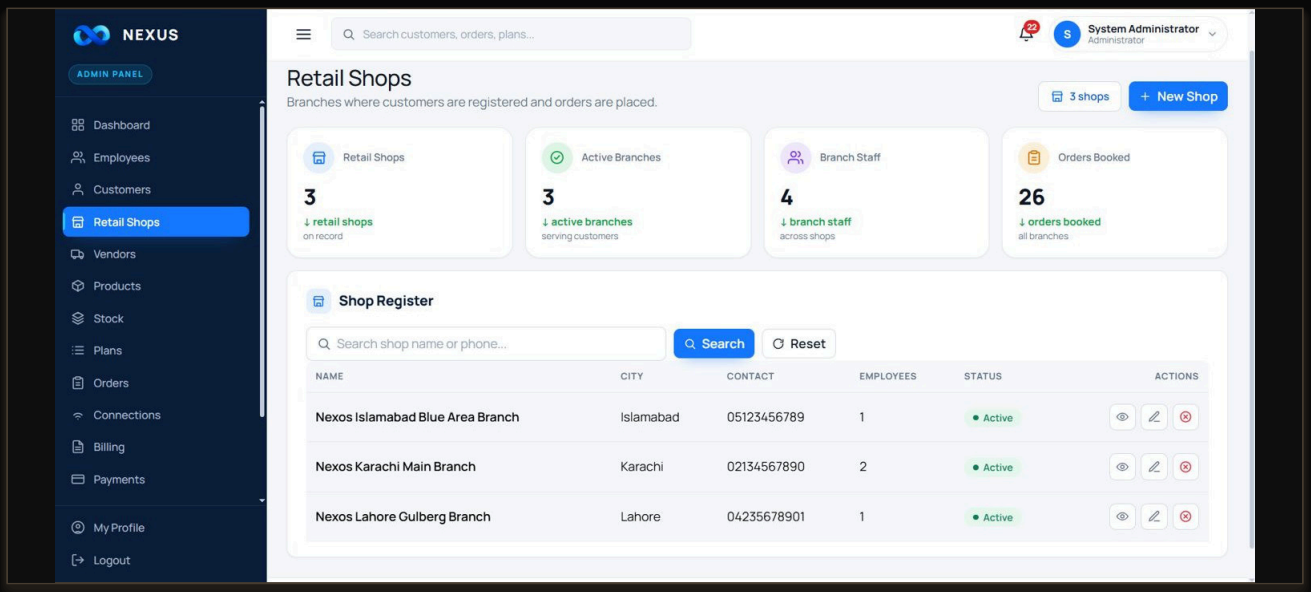


Figure 26: Retail Shops

Purpose: This page maintains the retail branches (Islamabad Blue Area, Karachi Main and Lahore Gulberg) with city, contact number, number of employees and status. The city of a shop decides which customers it serves and which city code is used in new account numbers.

Products

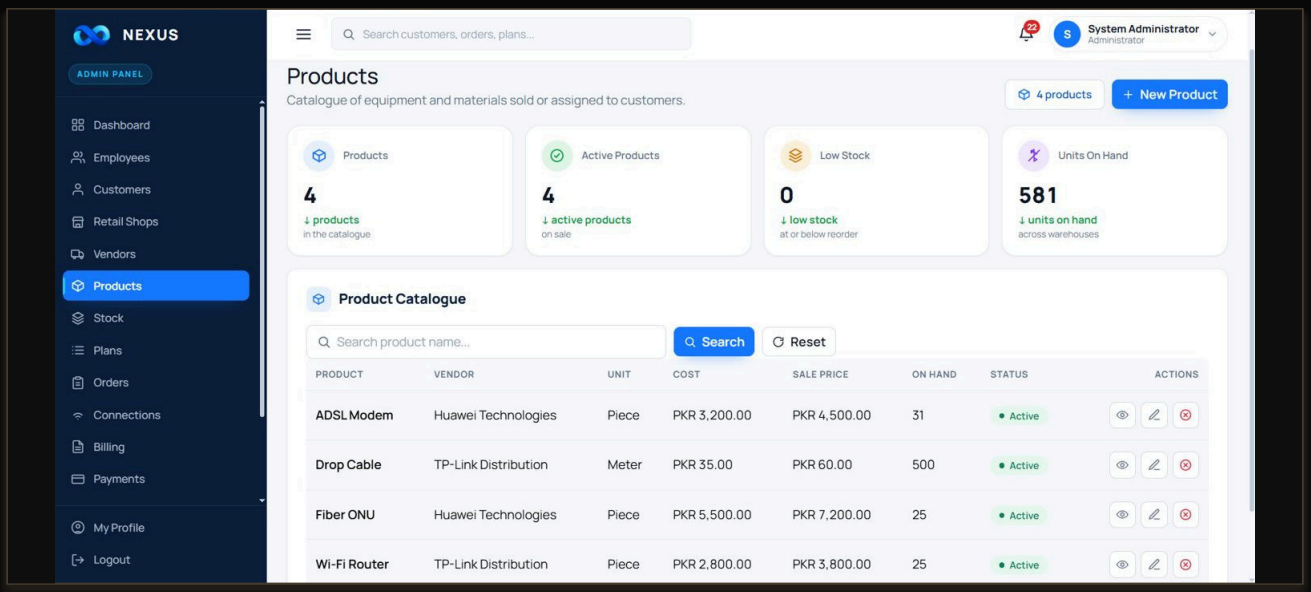


Figure 27: Products

Purpose: The product catalogue lists the equipment and materials sold or assigned to customers, such as ADSL modems, drop cable, fiber ONUs and Wi-Fi routers, with vendor, unit, cost price, sale price, quantity on hand and status. Creating a product also creates its stock record with an opening quantity.

Vendors

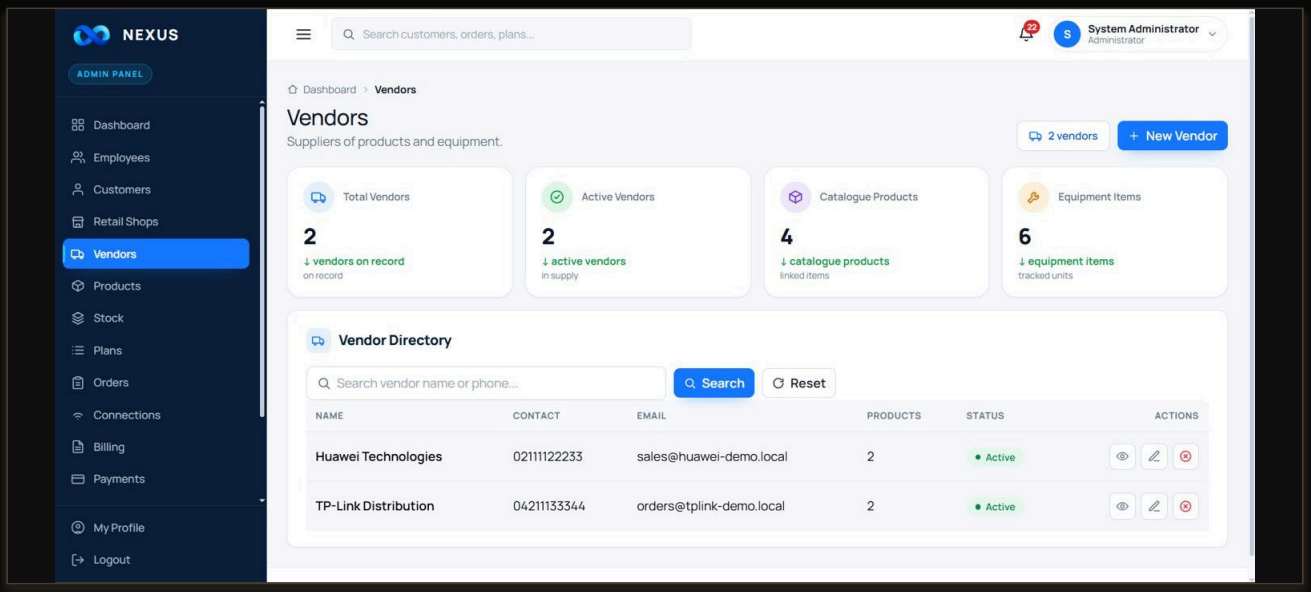


Figure 28: Vendors

Purpose: Vendors are the suppliers of equipment (here Huawei Technologies and TP-Link Distribution). The page shows contact details, email, number of products and status. A vendor that already has products or equipment is deactivated instead of being deleted.

Stock

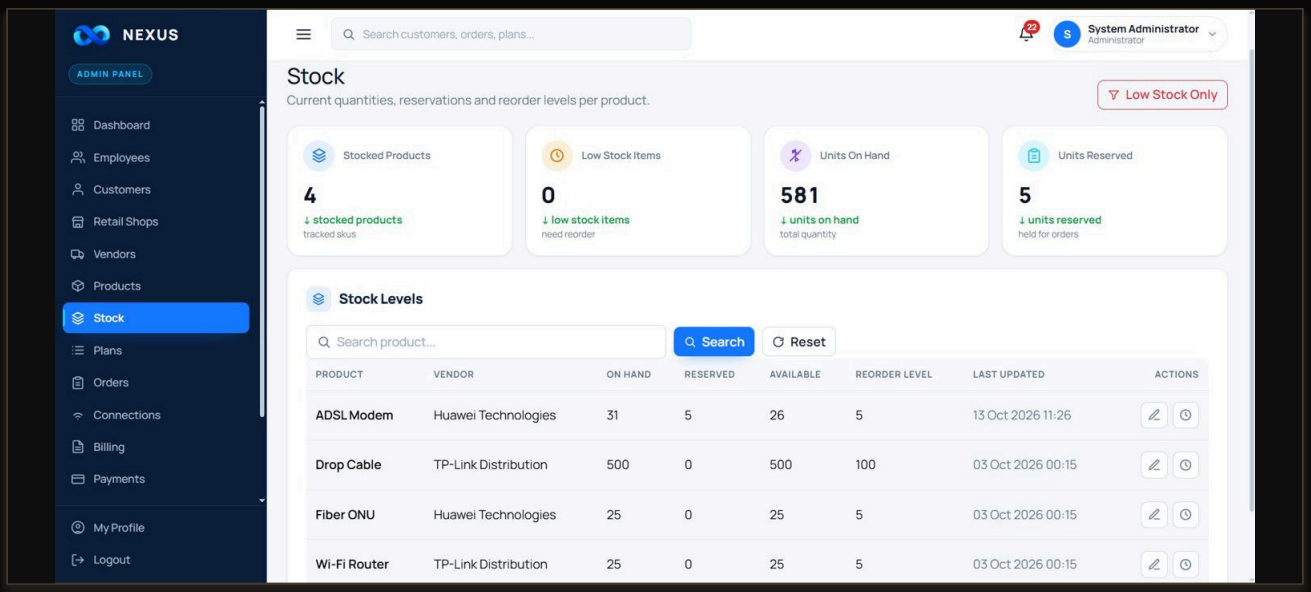


Figure 29: Stock

Purpose: The stock page shows quantity on hand, reserved quantity, available quantity and reorder level for each product, and highlights low stock. The admin can adjust stock or open the movement history, where every stock in, stock out and adjustment is recorded with reason, employee and time.

Plans

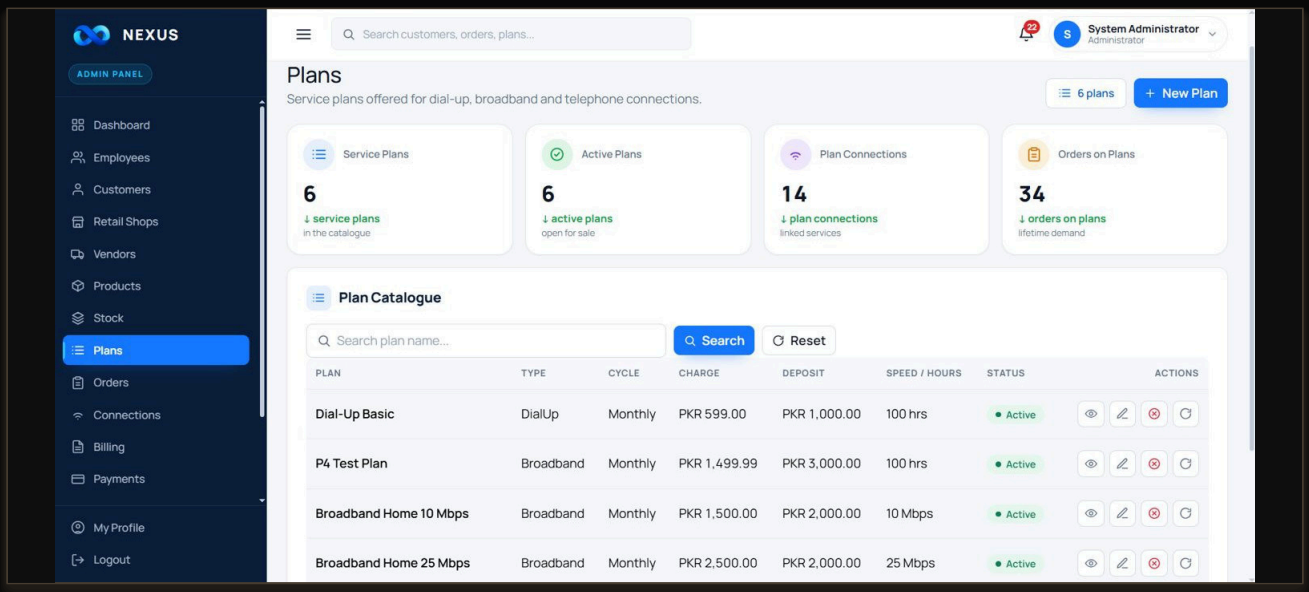


Figure 30: Plans

Purpose: Plans define what NEXUS sells. For each plan the admin sets connection type, billing cycle, speed or free hours, monthly charge, security deposit and installation charge. A plan can be activated, deactivated, edited or deleted (if unused); inactive plans disappear from the website and order forms.

Orders

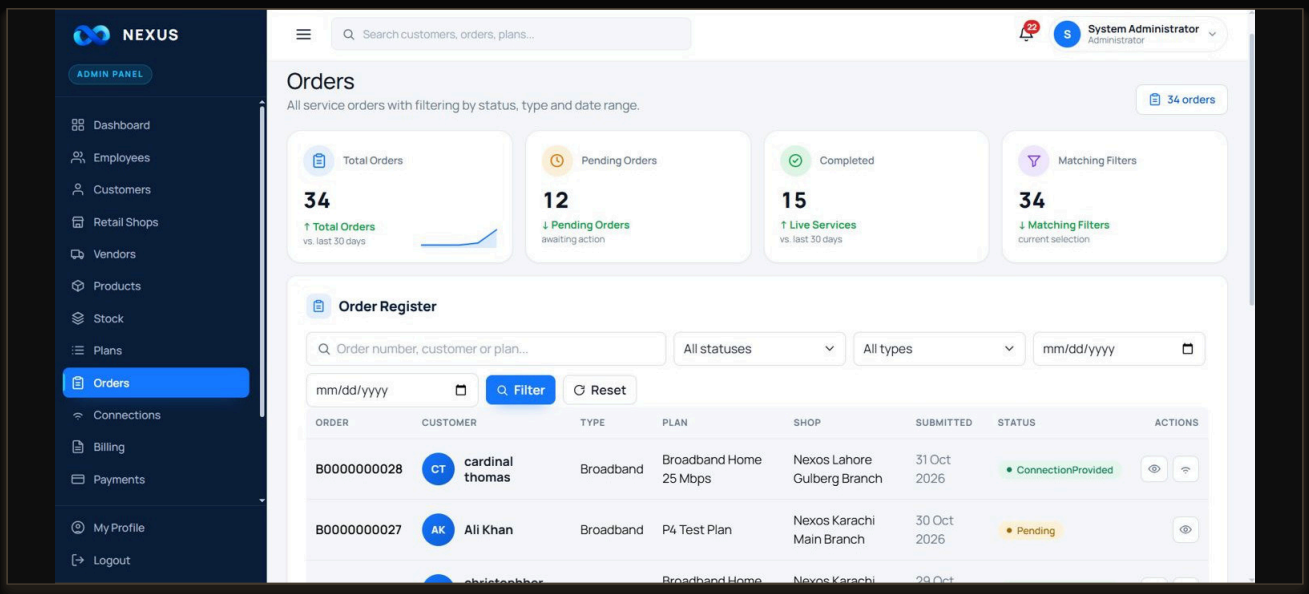


Figure 31: Orders

Purpose: The order register shows all orders of all branches with filters for status, connection type and date range. The admin can open any order, change its status through the validated workflow and send an “order received” email to the customer. Summary cards show total, pending and completed orders.

Connections

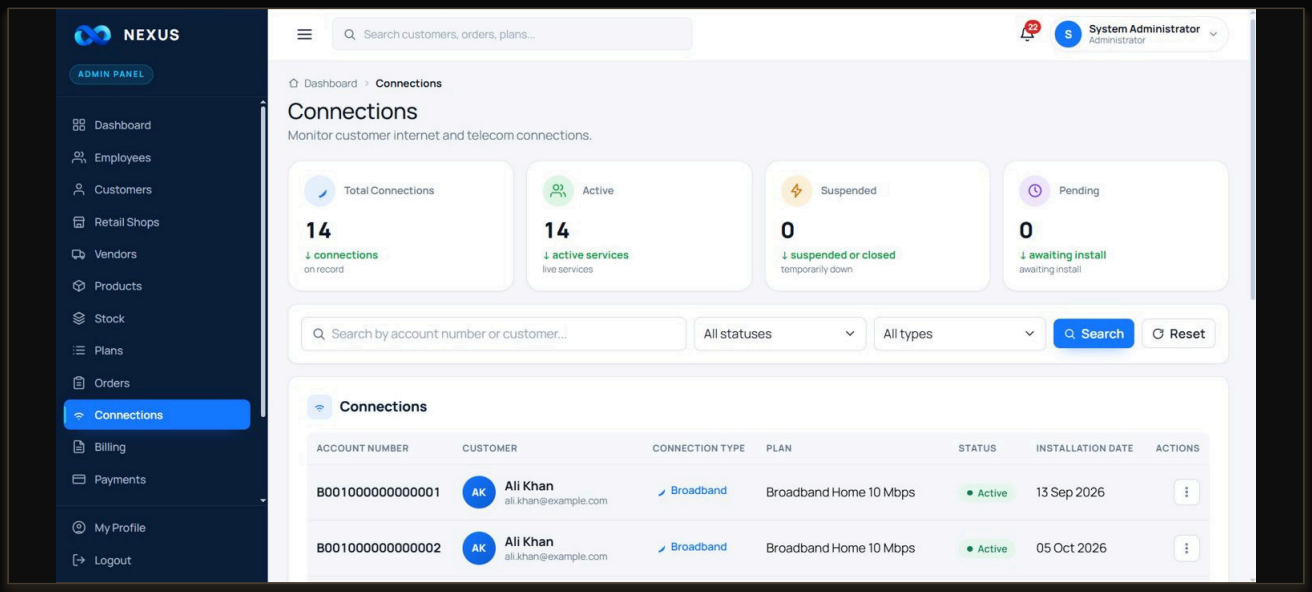


Figure 32: Connections

Purpose: This page monitors every customer connection with account number, customer, type, plan, status and installation date, and shows totals for active, suspended and pending connections. The admin can open a connection and change its status; a permanently inactive or cancelled connection can never be reactivated.

Billing

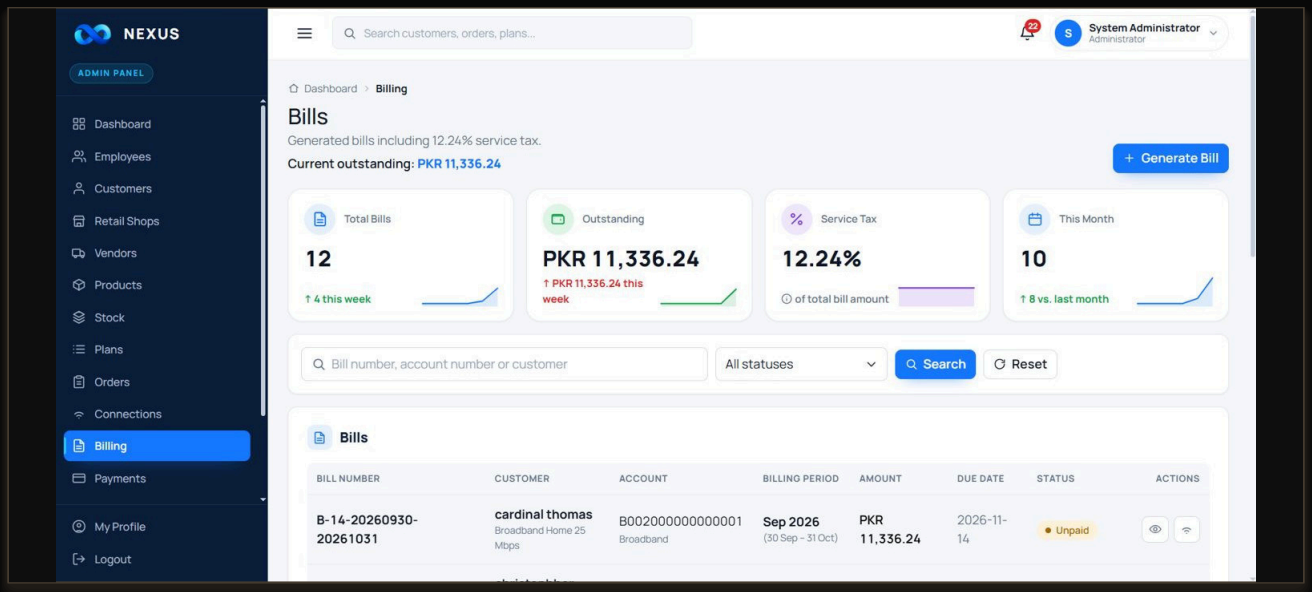


Figure 33: Billing

Purpose: The billing page lists all generated bills with billing period, amount, due date and status, and shows total bills, outstanding amount, the 12.24% service tax rate and bills generated this month. “Generate Bill” creates a new bill using the central billing calculation.

Payments

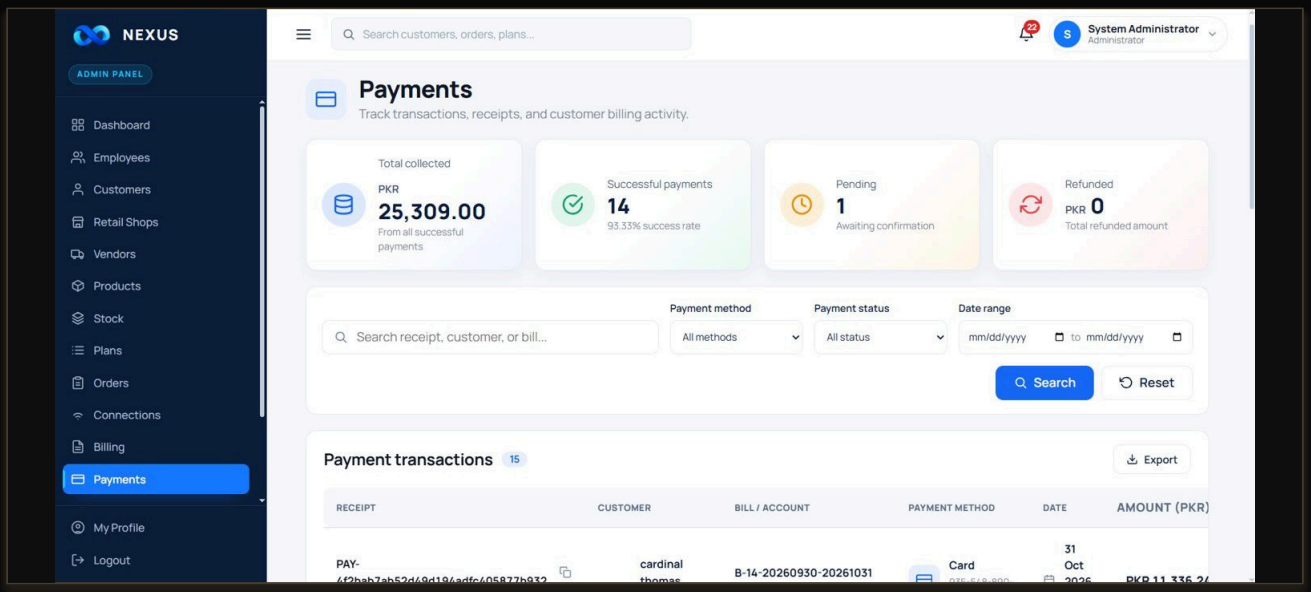


Figure 34: Payments

> **Purpose:** The payments page tracks every transaction with receipt number, customer, bill, method, date and amount. Cards show total collected, successful payments, pending payments and refunds; filters narrow the list by method, status and date, and Export downloads the result as a CSV file.

Admin Profile

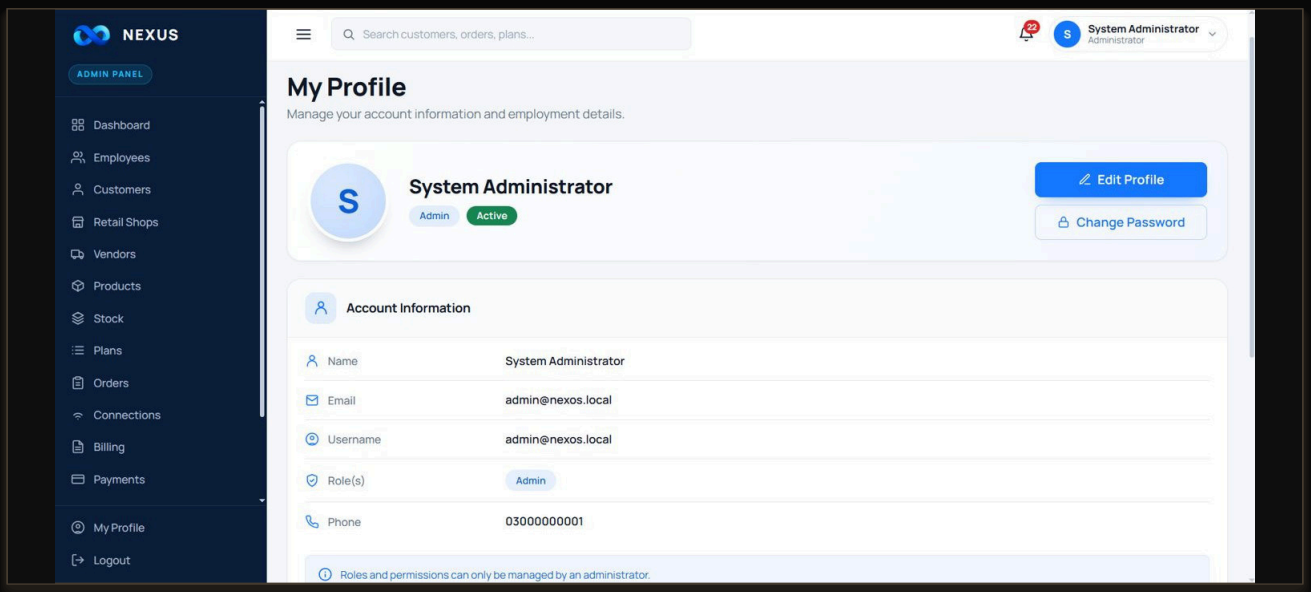


Figure 35: Admin Profile

> **Purpose:** The admin's own account page with name, email, username, role and phone number, plus Edit Profile and Change Password buttons. Roles and permissions can only be changed by an administrator.

Retail Panel

The Retail Panel is used by branch staff. A retail employee registers customers of his own city, places and tracks service orders, verifies customer details and forwards orders to the technical team. Bills and payments are shown for reference only, because billing is handled by the accounts team. Every list is limited to the logged-in employee's branch.

Retail Dashboard

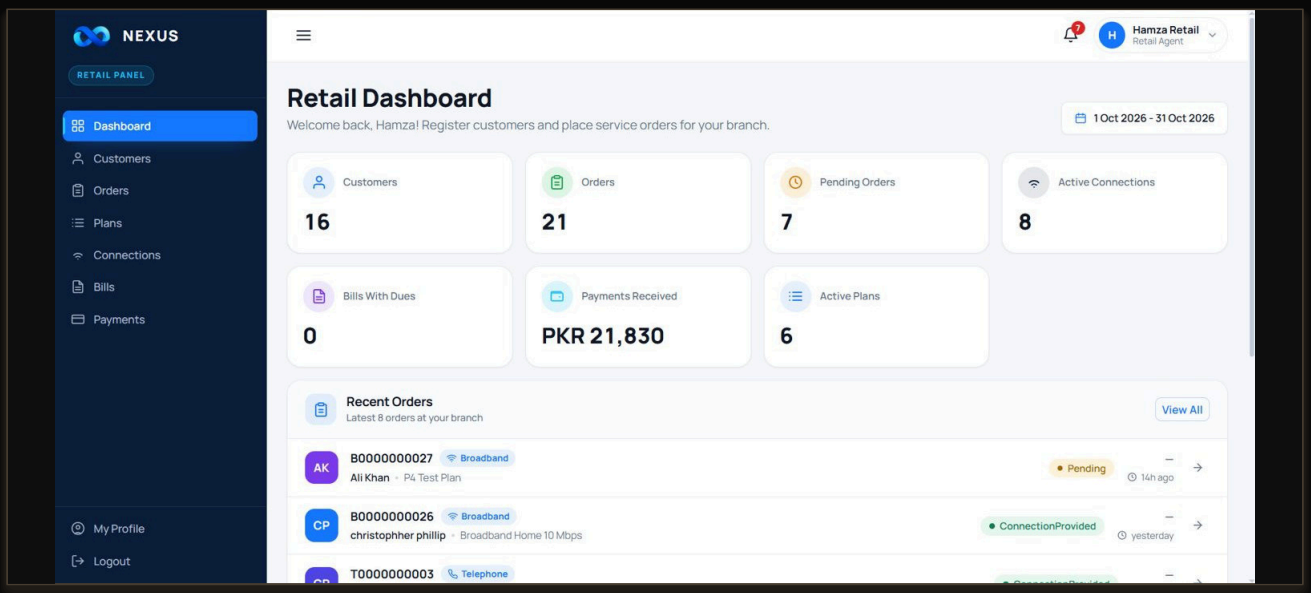


Figure 36: Retail Dashboard

Purpose: The branch dashboard shows customers, orders, pending orders and active connections, followed by bills with dues, payments received and active plans. "Recent Orders" lists the latest eight orders of the branch with their status, so staff can see at once which orders still need action.

Retail Customers

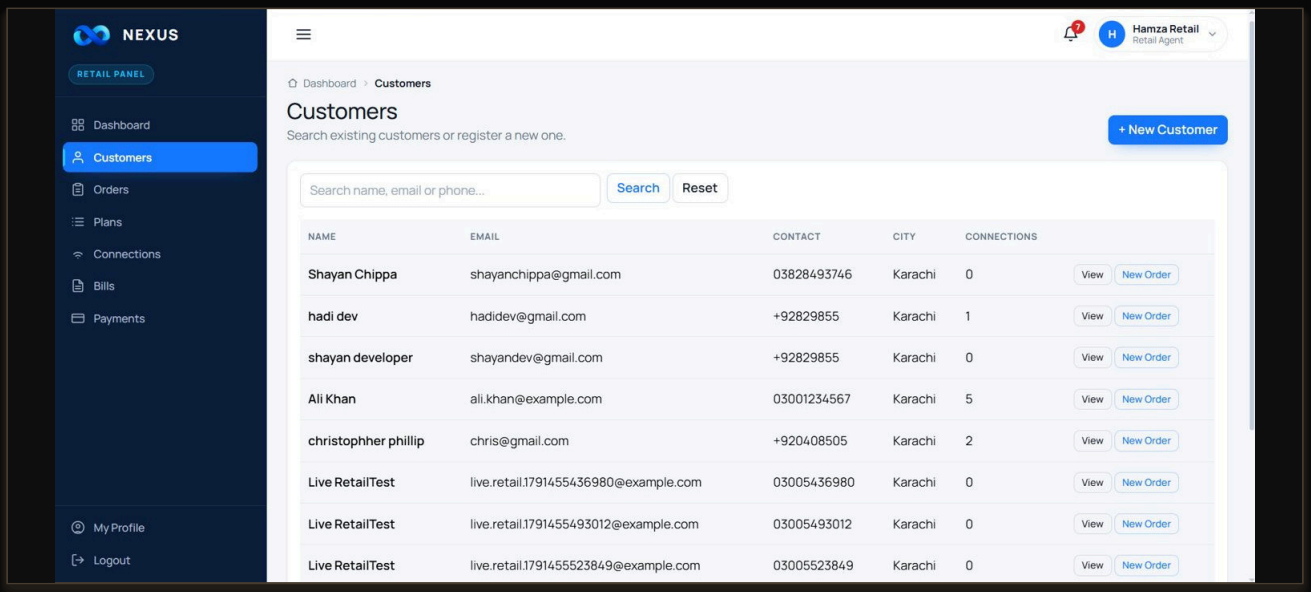


Figure 37: Retail Customers

Purpose: Staff can search customers by name, email or phone and see city and number of connections. “+ New Customer” registers a new customer in the shop’s city, and the New Order button on each row starts an order for that customer.

Retail Orders

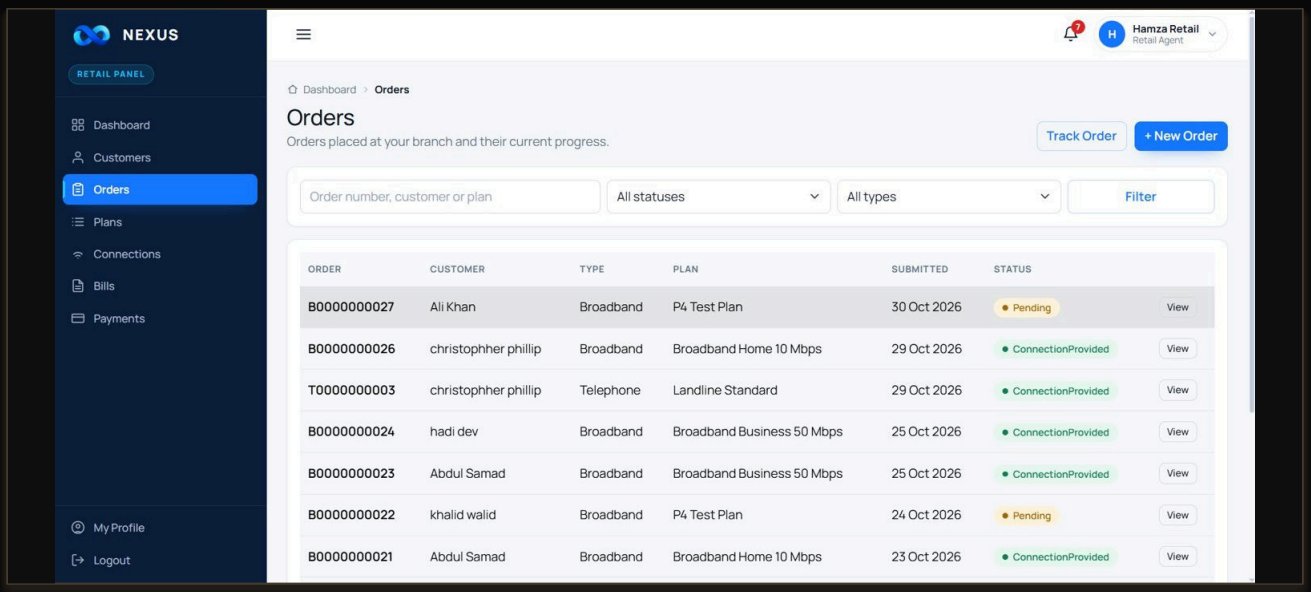


Figure 38: Retail Orders

Purpose: This page lists the orders placed at the branch with order number, customer, type, plan, date and status. Filters by order number, status and type help find an order quickly, and View opens the order for verification or details. Pending orders are the ones waiting for retail action.

New Order

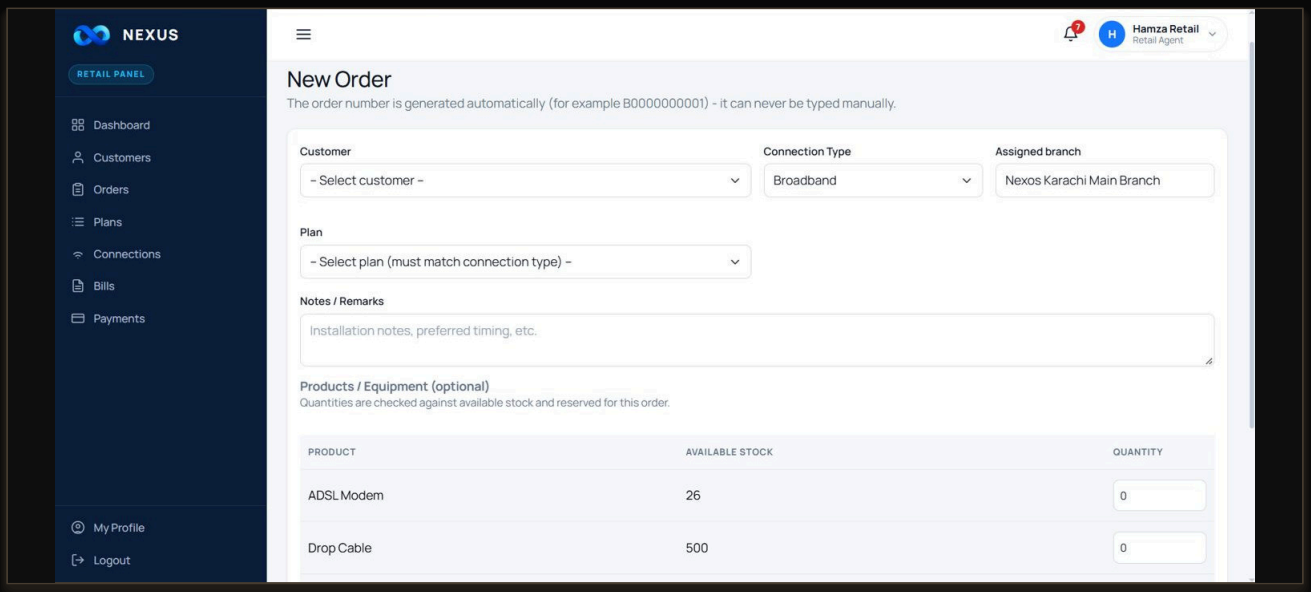


Figure 39: New Order

Purpose: Retail staff create an order for a customer by choosing the customer, connection type and plan (which must match the type). Optional products or equipment can be reserved here; the available stock is shown and checked. The order number is generated automatically and can never be typed by hand.

Track Order

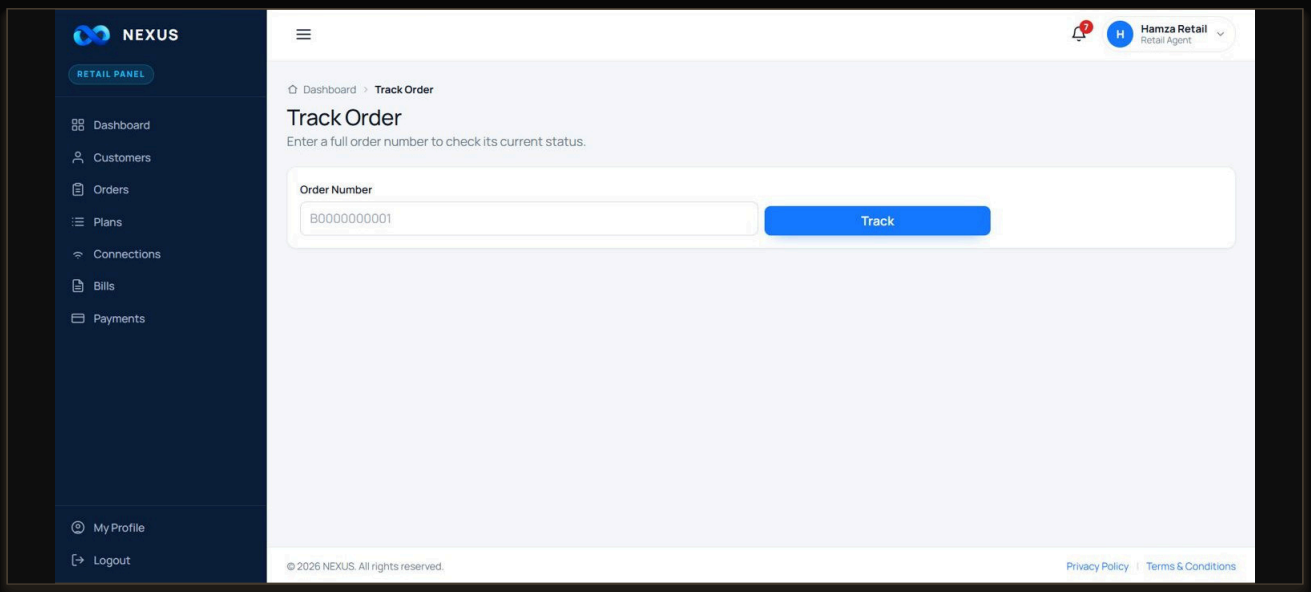


Figure 40: Track Order

Purpose: A quick lookup: the employee enters a full order number (for example B0000000001) and presses Track to see its current status without searching the whole list.

Available Plans

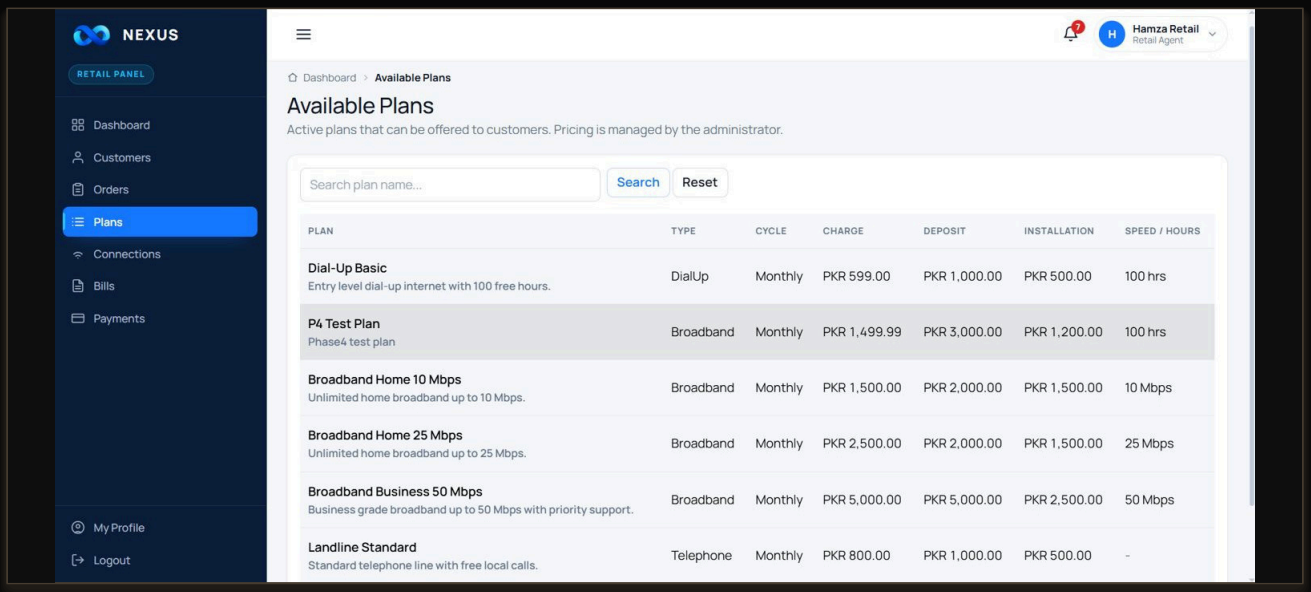


Figure 41: Available Plans

> **Purpose:** A read-only list of the active plans that can be offered to customers, with type, cycle, monthly charge, deposit, installation charge and speed or hours. Prices are managed by the administrator only.

Retail Connections

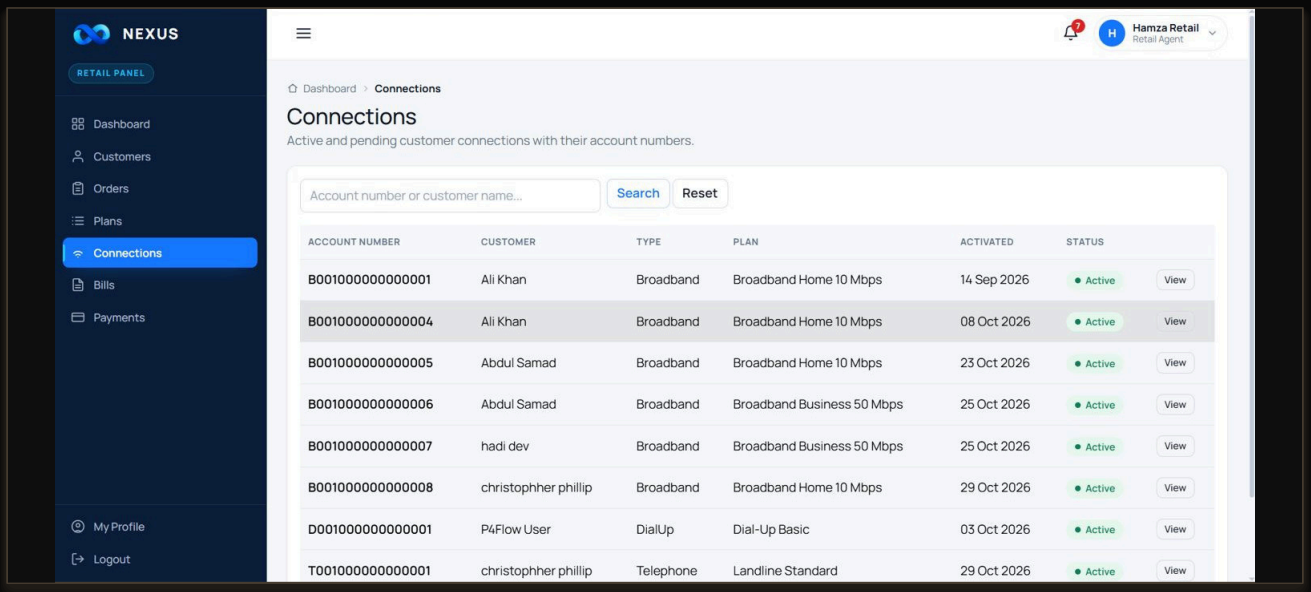


Figure 42: Retail Connections

> **Purpose:** Shows the active and pending connections of the branch's customers with account number, customer, type, plan, activation date and status, so staff can answer customer questions about their service.

Retail Bills

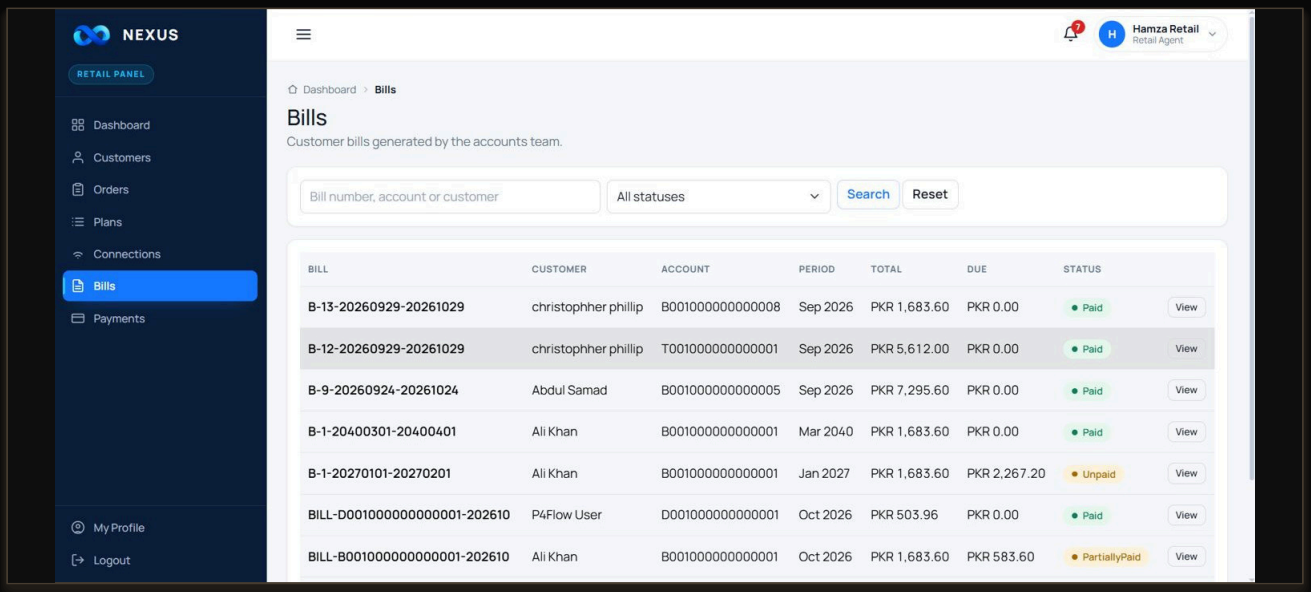


Figure 43: Retail Bills

Purpose: Lists the bills generated by the accounts team for the branch's customers with bill number, account, period, total, due and status (Paid, Unpaid, PartiallyPaid). Retail staff can view bills but not create or change them.

Retail Payments

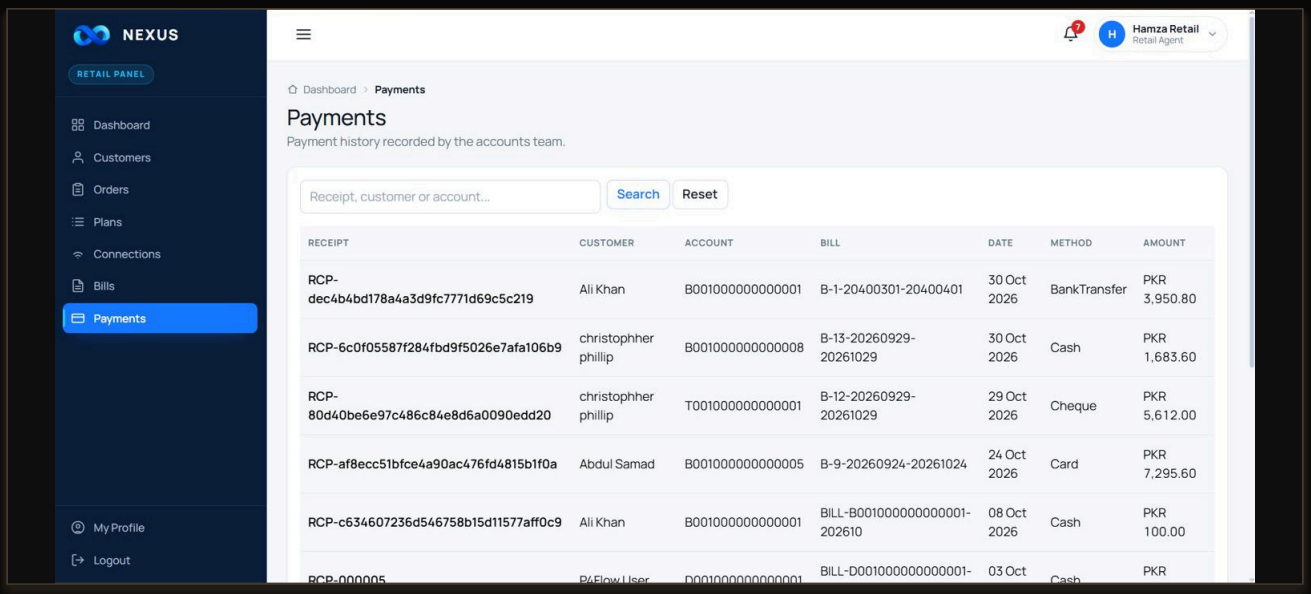


Figure 44: Retail Payments

Purpose: Shows the payment history recorded by the accounts team (receipt, customer, account, bill, date, method and amount) so branch staff can confirm to a customer that a payment was received.

Retail Profile

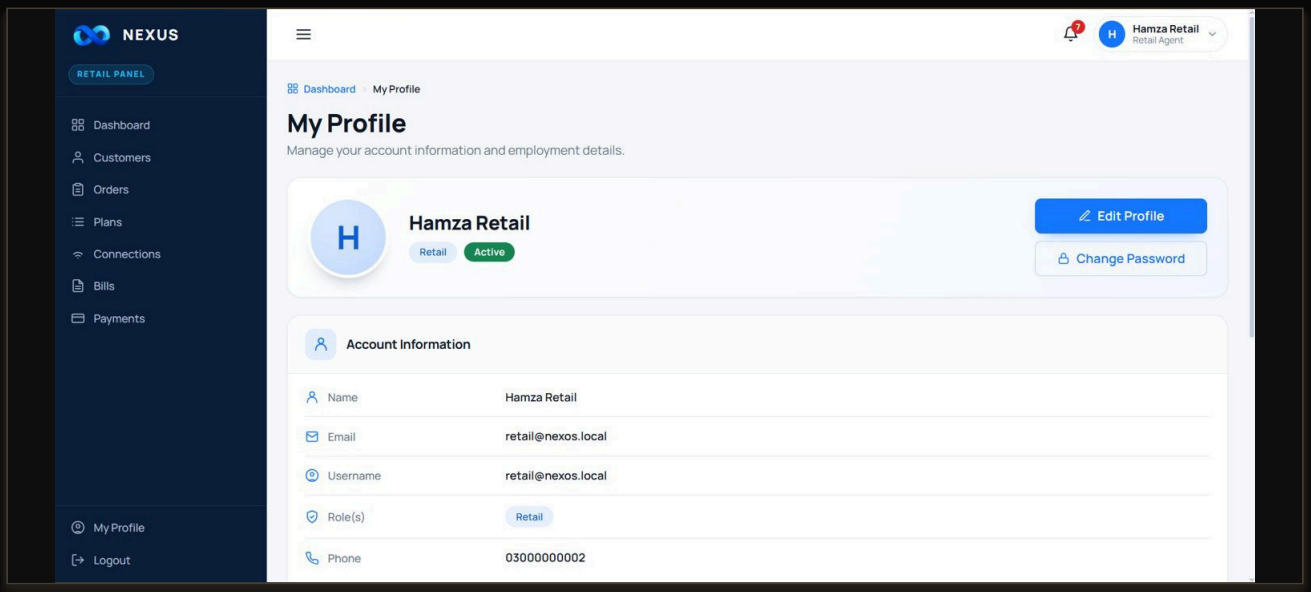


Figure 45: Retail Profile

Purpose: The retail employee's own account page showing name, email, username, role and phone, with options to edit the profile and change the password.

Accounts Panel

The Accounts Panel handles all money matters. Accounts staff generate bills, record cash, cheque and card payments, verify or reject payments submitted online by customers, follow up outstanding dues with reminders, and look up customers and connections for billing.

Accounts Dashboard

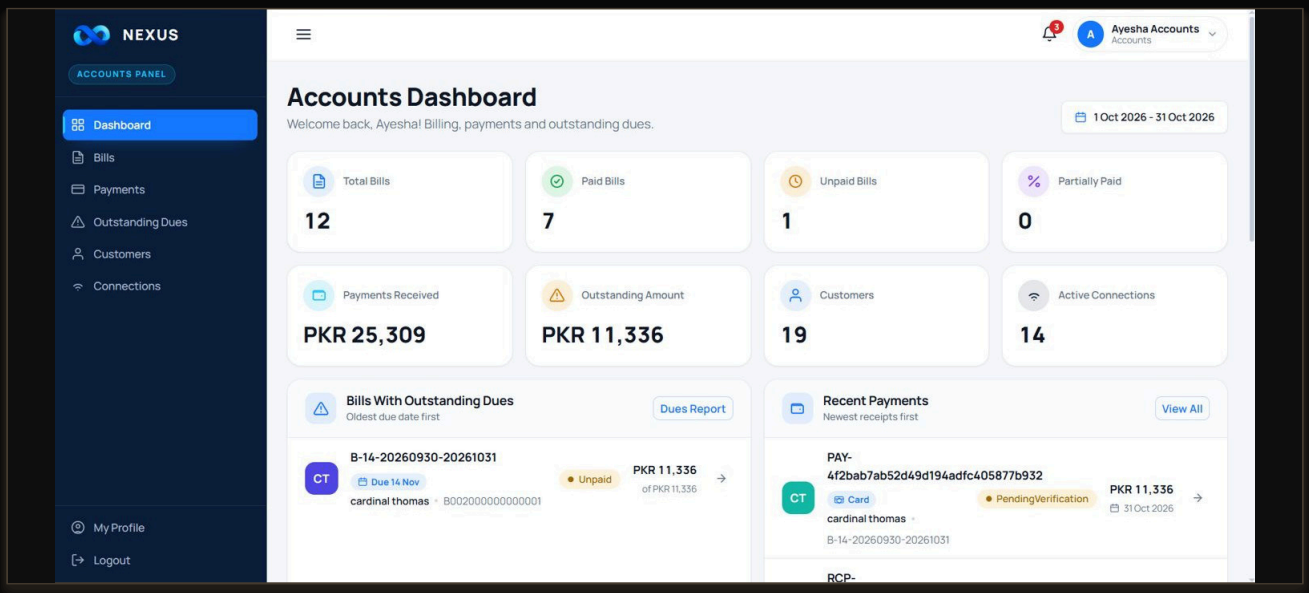


Figure 46: Accounts Dashboard

Purpose: The financial overview shows total, paid, unpaid and partially paid bills, payments received, outstanding amount, customers and active connections. Below, “Bills With Outstanding Dues” lists the oldest unpaid bills first and “Recent Payments” lists the newest receipts, including payments still waiting for verification.

Accounts - Bills

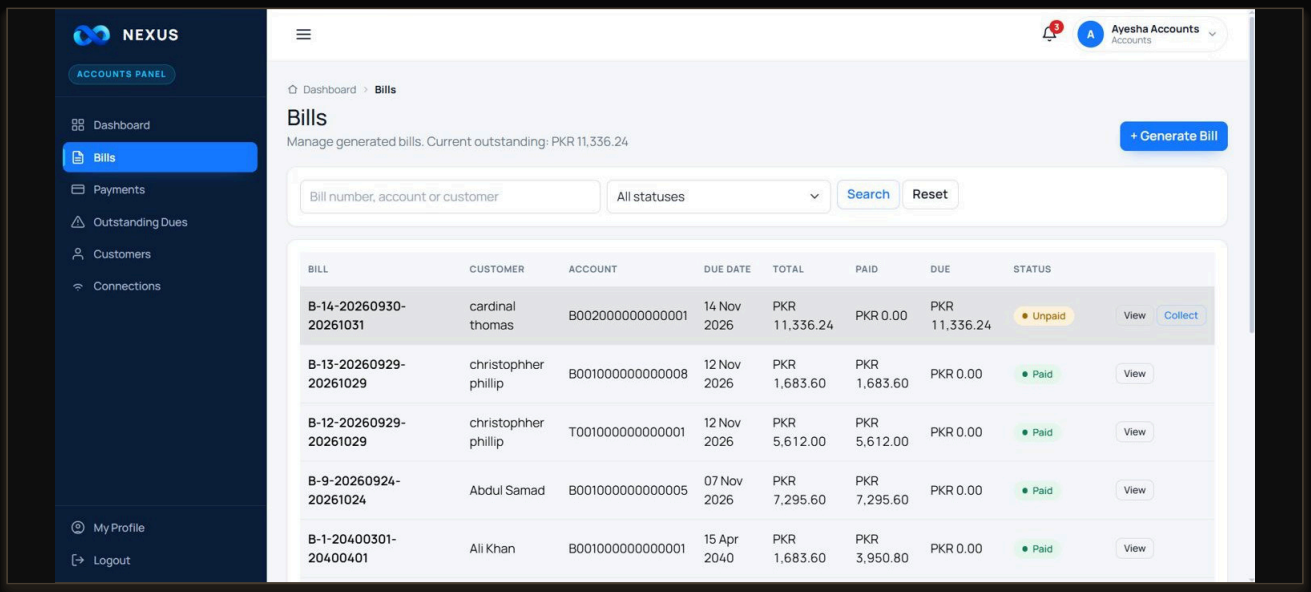


Figure 47: Accounts - Bills

Purpose: The bill register shows bill number, customer, account, due date, total, paid, due and status, with the current outstanding amount at the top. “+ Generate Bill” creates a bill, View opens it, and Collect on an unpaid bill jumps straight to recording a payment.

Generate Bill

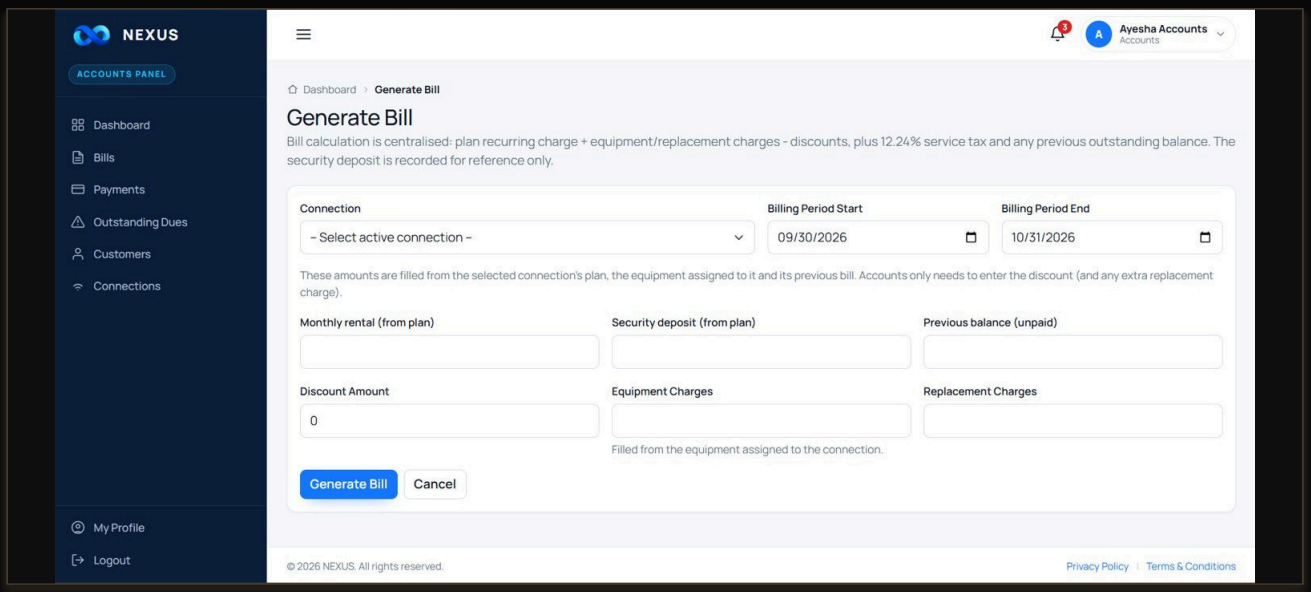


Figure 48: Generate Bill

Purpose: The form for creating a bill. The accounts officer selects an active connection and billing period; the monthly rental, security deposit, previous balance and equipment charges are filled automatically from the plan, assigned equipment and previous bill. Only the discount (and any extra replacement charge) is entered. The system adds 12.24% service tax to produce the total.

Accounts - Payments

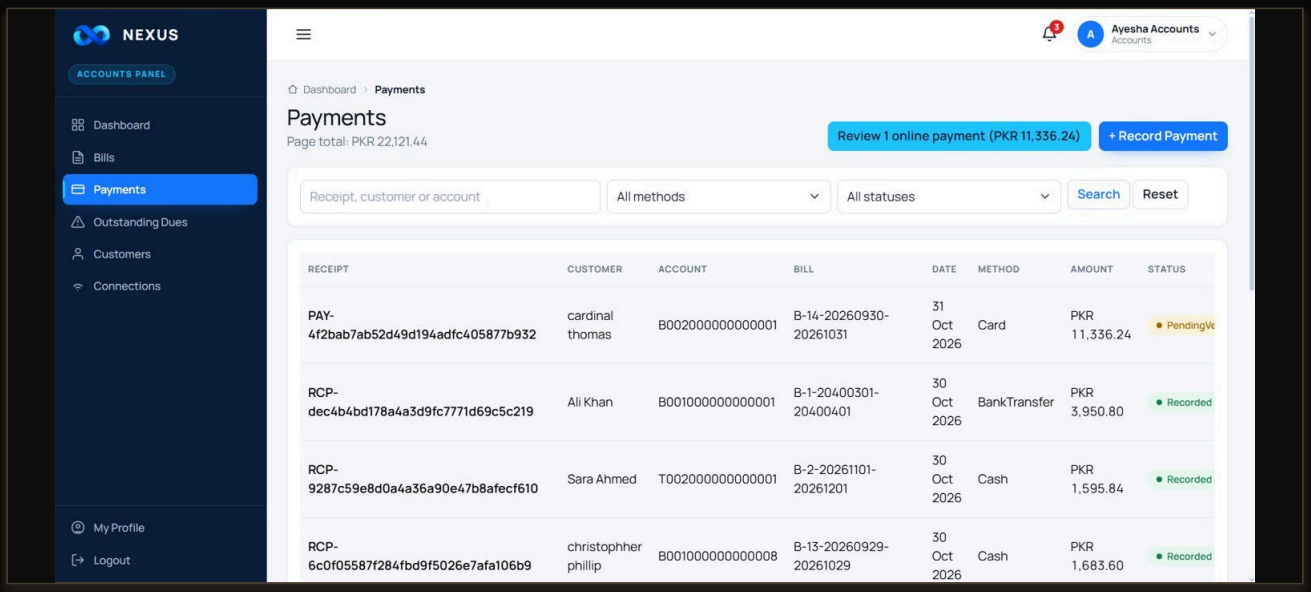


Figure 49: Accounts - Payments

Purpose: All payment records with receipt, customer, account, bill, date, method, amount and status. “+ Record Payment” enters a cash, card, bank or cheque payment received at the counter, and “Review online payment” opens the payments submitted by customers, which the officer verifies or rejects. Only a verified payment updates the bill.

Outstanding Dues

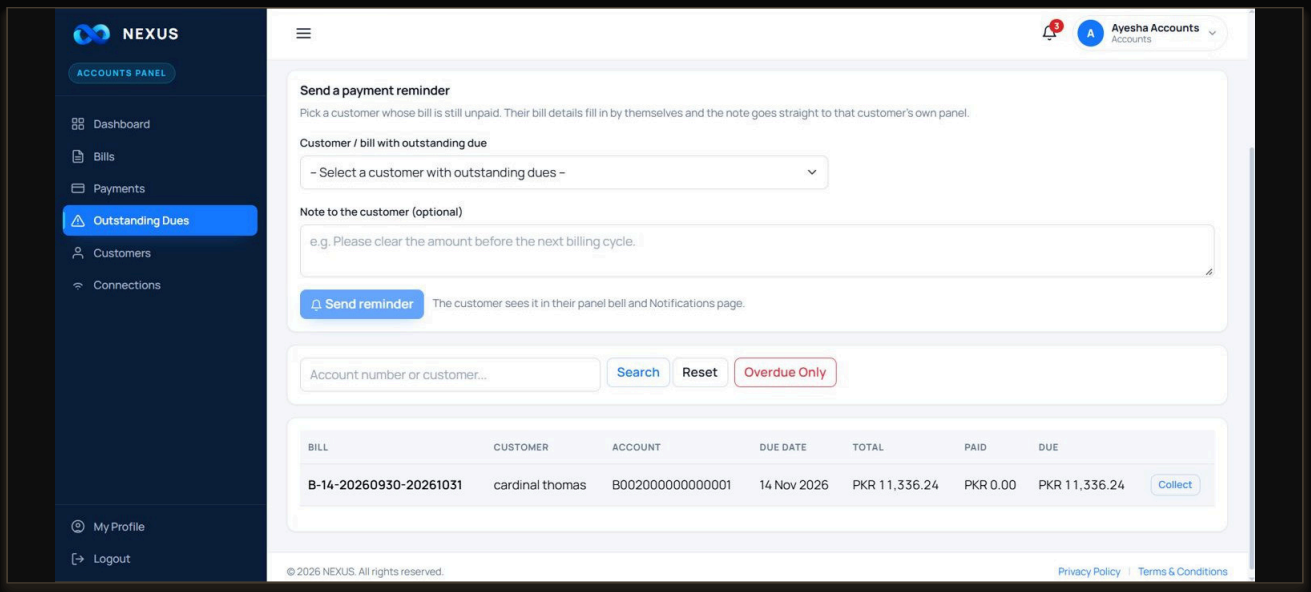


Figure 50: Outstanding Dues

Purpose: Lists all bills that are still unpaid, with an “Overdue Only” filter. At the top, a reminder form lets accounts pick a customer bill and send a note that appears in the customer’s notification bell and Notifications page. Collect starts a payment entry for the bill.

Accounts - Customers

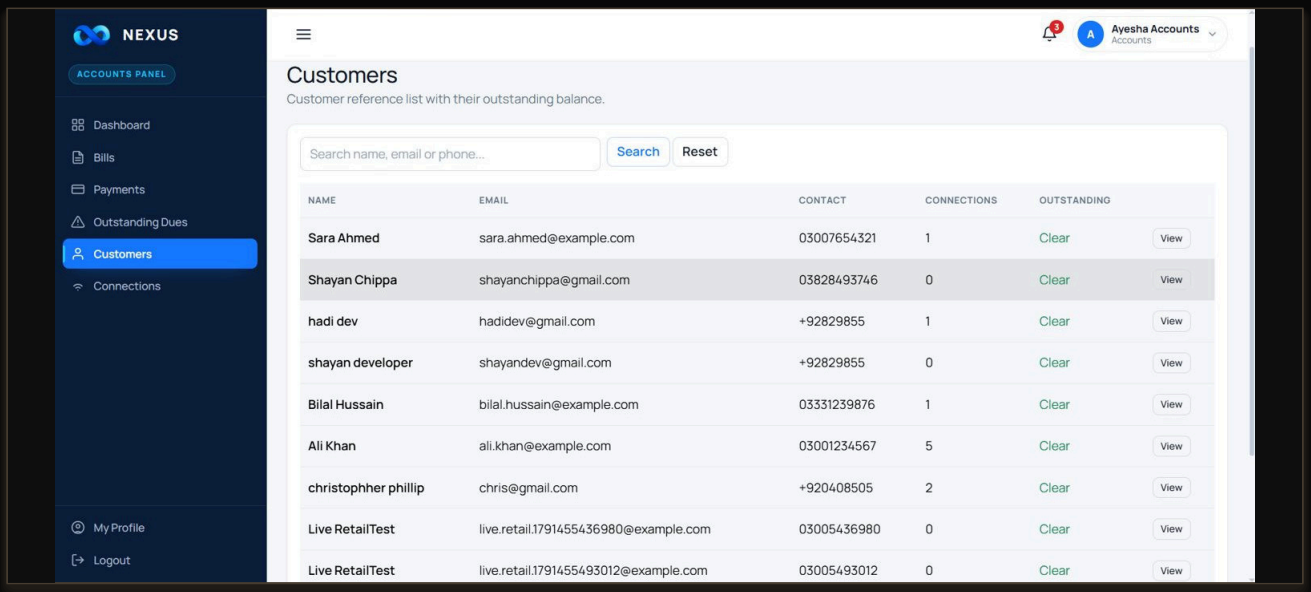


Figure 51: Accounts - Customers

Purpose: A customer reference list with name, email, contact, number of connections and outstanding balance (“Clear” when nothing is due), so the accounts team can check a customer’s financial position quickly.

Accounts - Connections

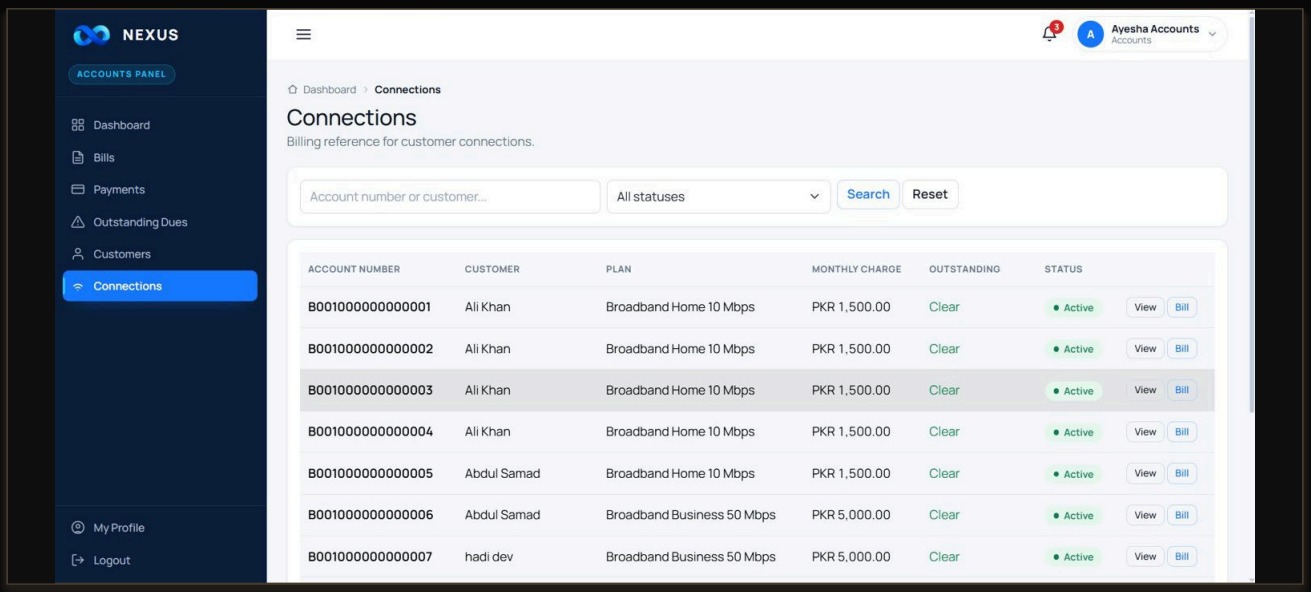


Figure 52: Accounts - Connections

Purpose: Billing reference for every connection: account number, customer, plan, monthly charge, outstanding amount and status. The Bill button opens the bill-generation form for that connection.

Accounts Profile

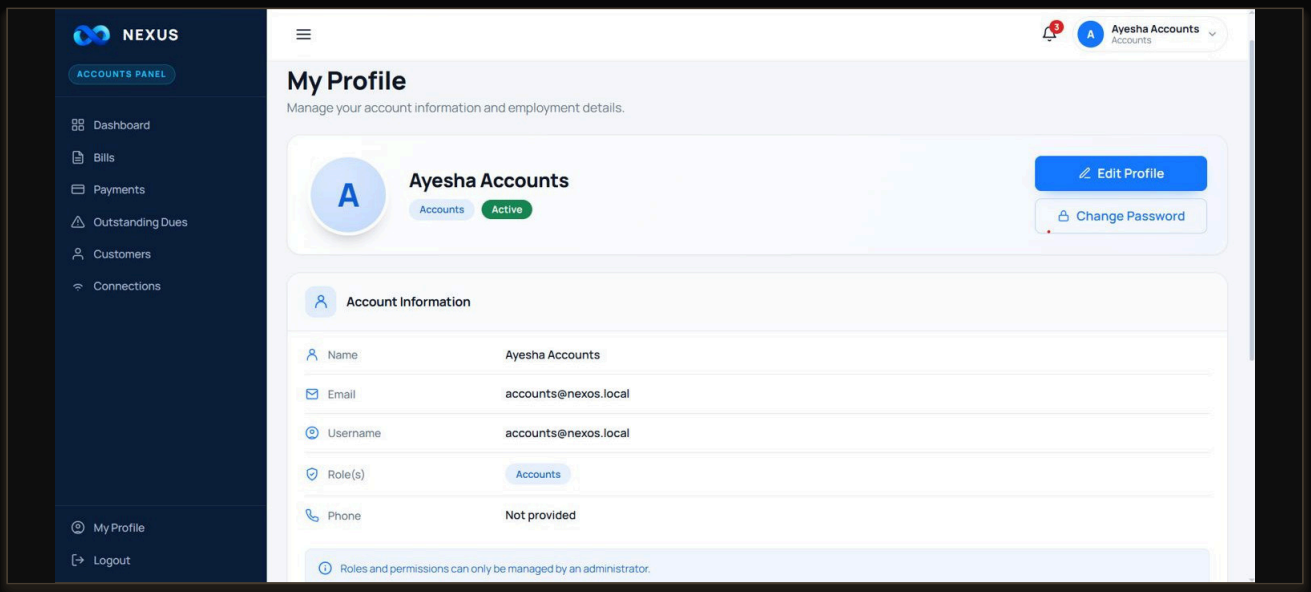


Figure 53: Accounts Profile

> **Purpose:** The accounts employee's own account page with name, email, username, role and phone, plus Edit Profile and Change Password.

Technical Panel

The Technical Panel is used by the field and network team. After retail verifies an order, technical staff record the feasibility (site check), create and activate the connection, assign equipment from stock and handle returns and repairs. Customer requests to cancel or return equipment are also reviewed here.

Technical Dashboard

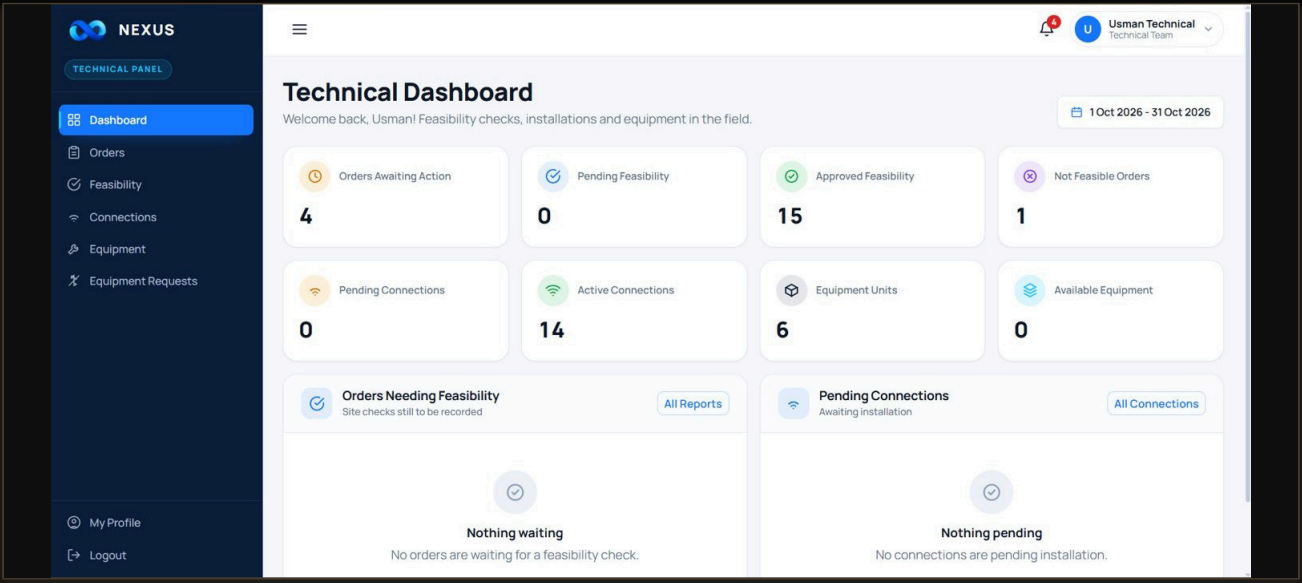


Figure 54: Technical Dashboard

Purpose: The dashboard shows orders awaiting action, pending and approved feasibility checks, not-feasible orders, pending and active connections, equipment units and available equipment. Two panels list the orders that still need a feasibility check and the connections awaiting installation.

Technical Orders

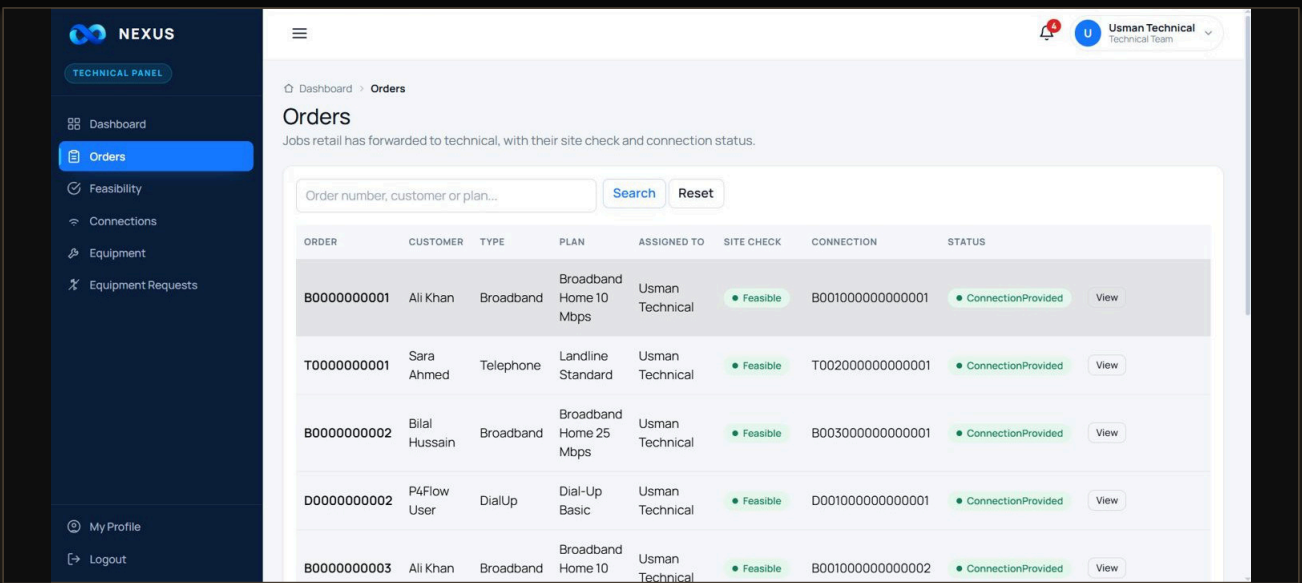


Figure 55: Technical Orders

Purpose: Lists the jobs that retail has forwarded to the technical team with customer, type, plan, assigned technician, site-check result, connection number and status. View opens the full order.

Technical Order Details

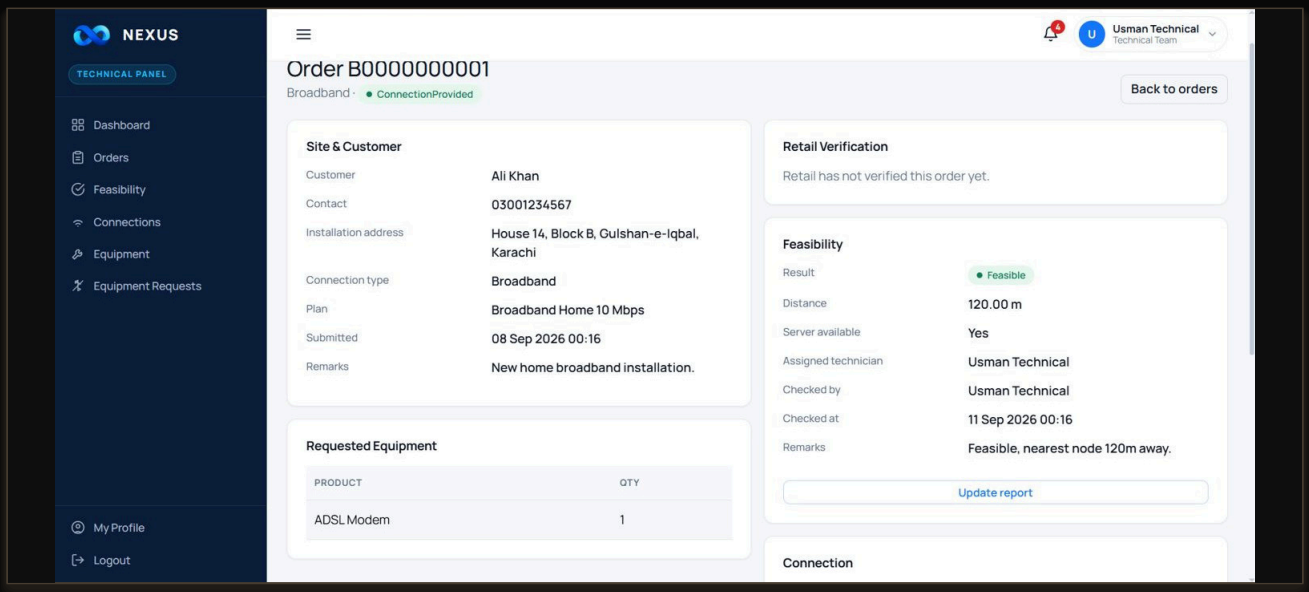


Figure 56: Technical Order Details

Purpose: The detailed view of one order: site and customer information with installation address, requested equipment, retail verification, the feasibility result (distance, server available, technician, date, remarks) with an “Update report” button, and the connection created from the order.

Feasibility Reports

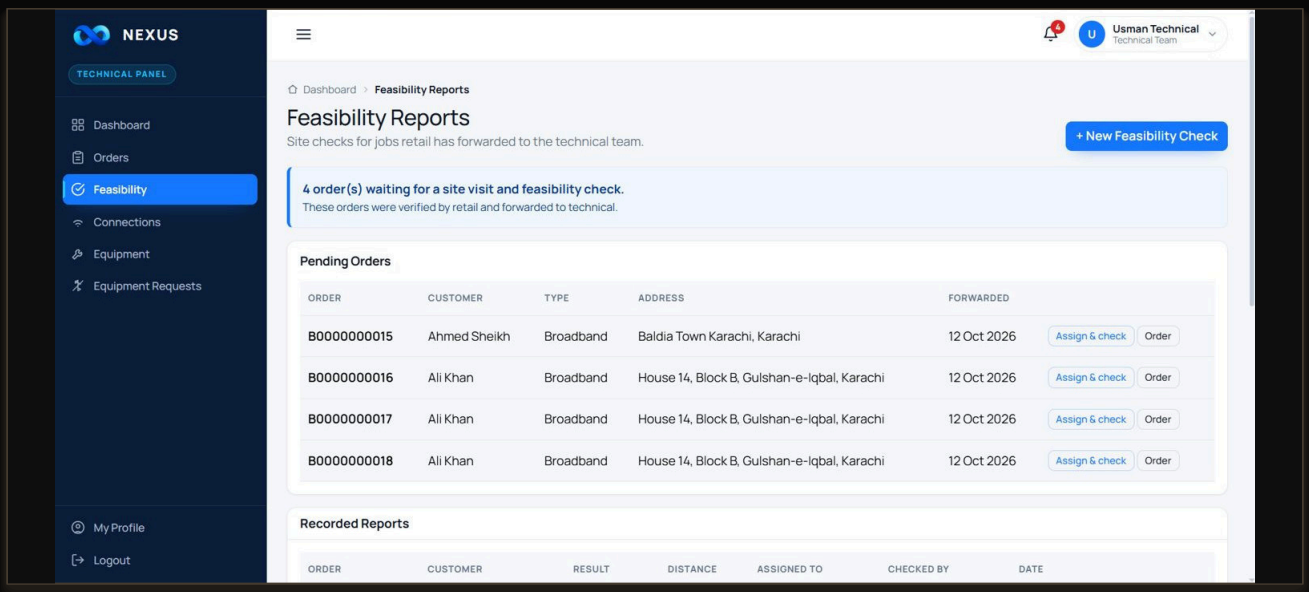


Figure 57: Feasibility Reports

Purpose: Shows orders waiting for a site visit and the reports already recorded. For each pending order the technician can press “Assign & check” to record the result. A banner reminds the team how many orders are waiting.

New Feasibility Check

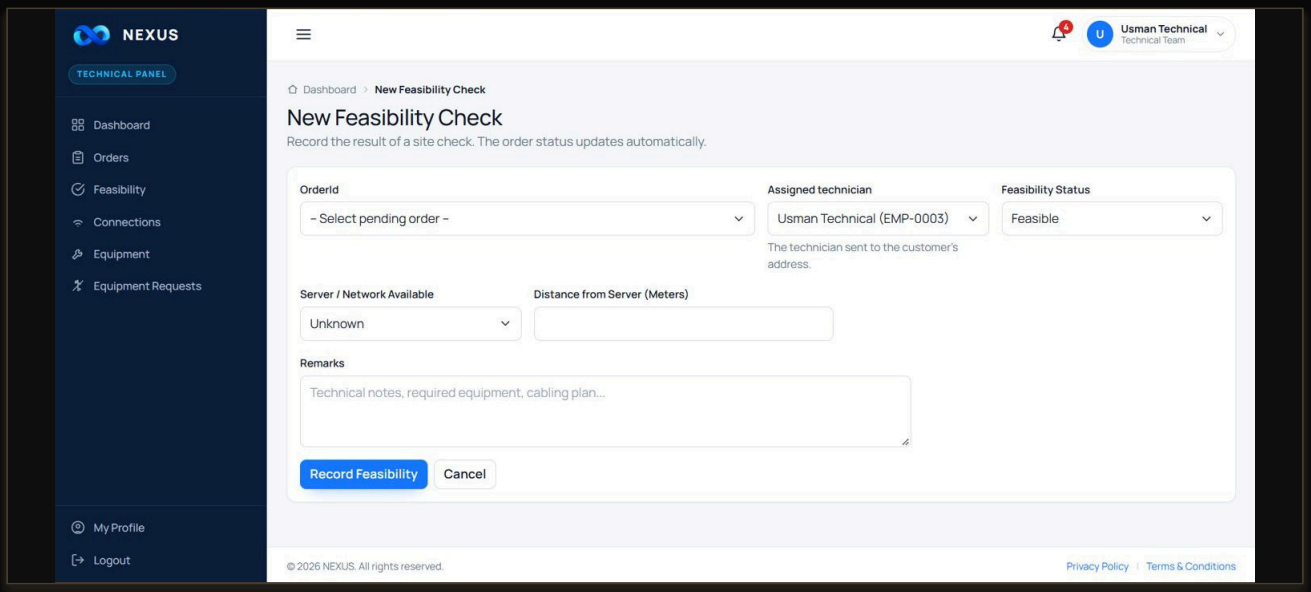


Figure 58: New Feasibility Check

Purpose: The form to record a site check: select the pending order, the assigned technician, the status (Feasible or Not Feasible), whether a server or network is available, the distance in metres and remarks. Saving the report updates the order status automatically.

Technical Connections

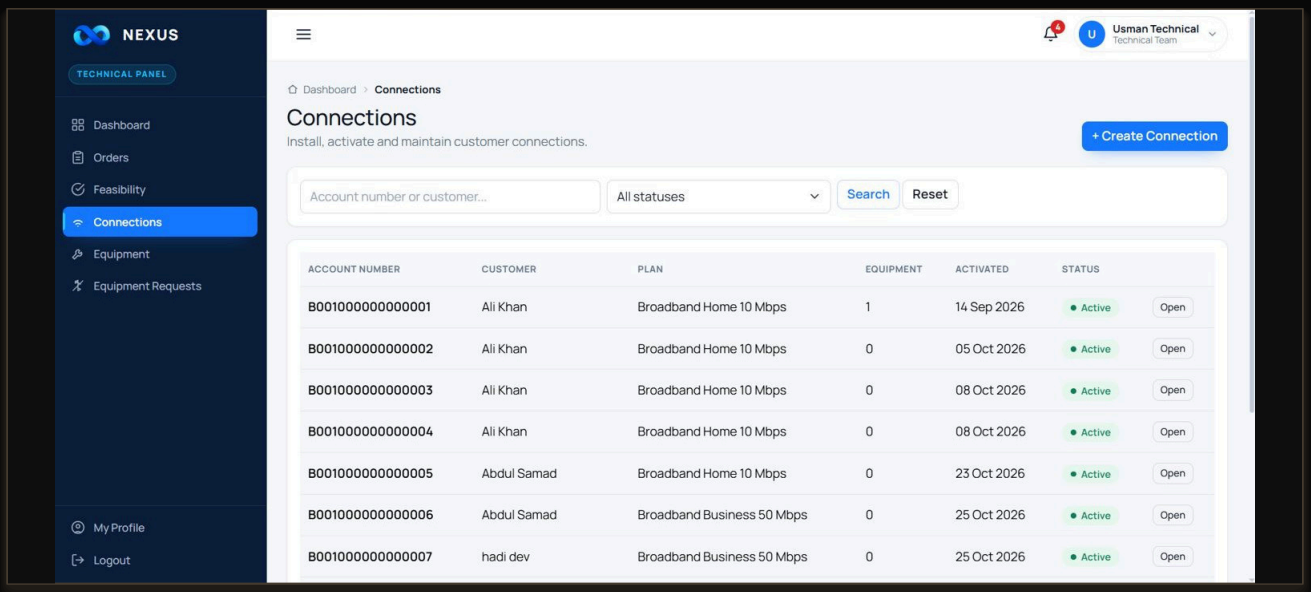


Figure 59: Technical Connections

Purpose: Lists customer connections with account number, customer, plan, number of equipment items, activation date and status. “+ Create Connection” makes a connection for a feasible order, and Open shows the connection to activate it or change its status.

Equipment Inventory

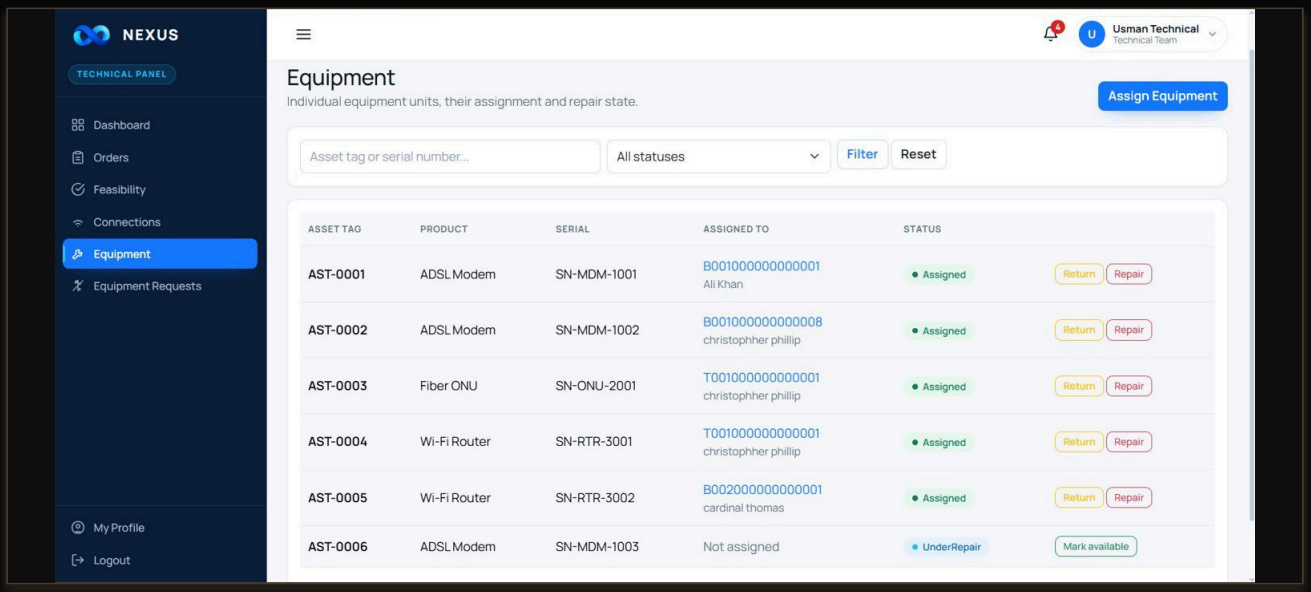


Figure 60: Equipment Inventory

> **Purpose:** Every individual equipment unit with asset tag, product, serial number, current connection and status. Technicians can assign a unit to a connection, return it to stock, send it for repair and mark it available again after repair.

Technical Profile

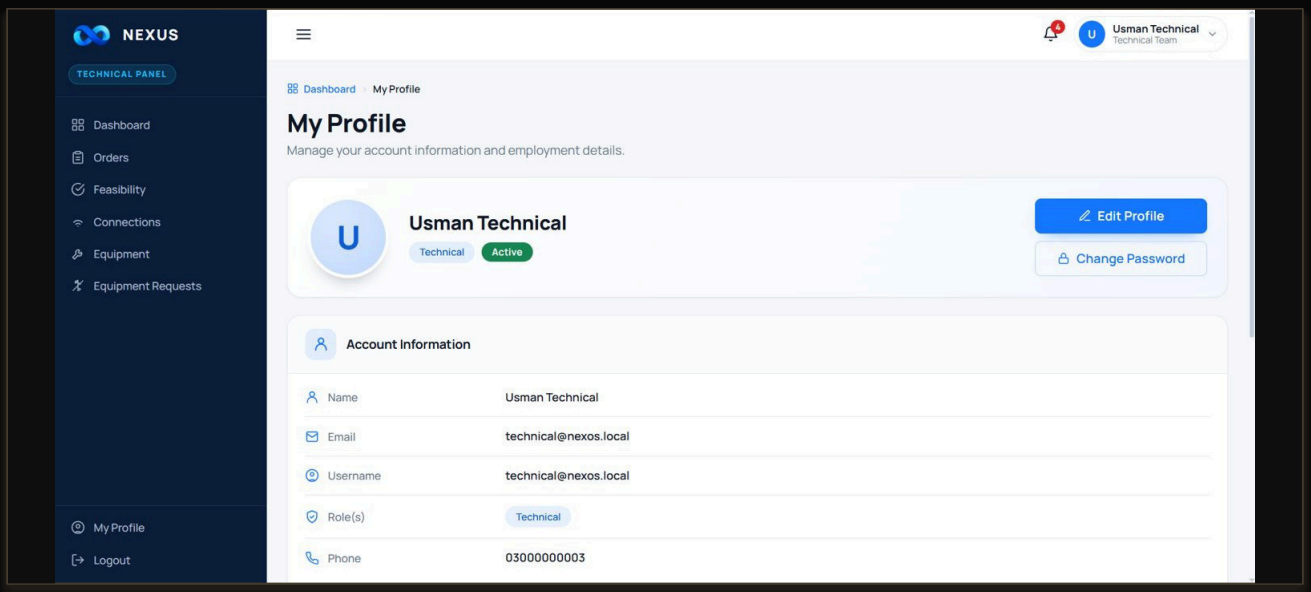


Figure 61: Technical Profile

> **Purpose:** The technical employee's own account page with name, email, username, role and phone, plus Edit Profile and Change Password.

Database Design

The application uses a Microsoft SQL Server database (NexosNetworkSystem) created with Entity Framework Core Code First migrations. It contains 26 application tables, shown in the image below, plus the standard ASP.NET Identity security tables (AspNetRoles, AspNetUserRoles, AspNetUserClaims, AspNetRoleClaims, AspNetUserLogins, AspNetUserTokens) and the migration history table. Each table has an Id primary key (PK); foreign keys (FK) connect the tables as shown in the ERD.

Database Tables and Columns

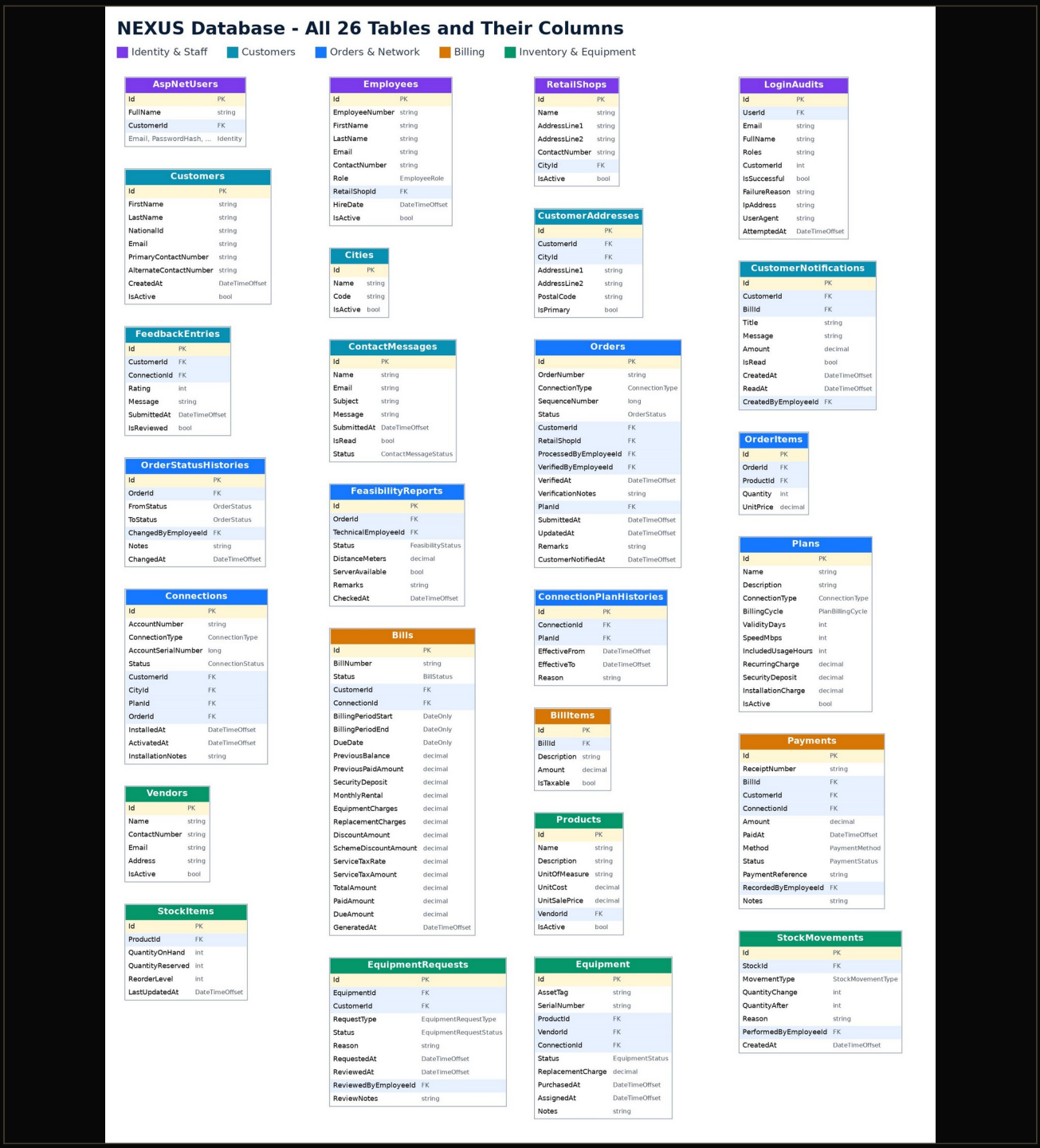


Figure 62: All tables of the NEXUS database with their columns

Purpose: This image shows every table and its columns. Table headers are coloured by module: purple for identity and staff, teal for customers, blue for orders and network, orange for billing and green for inventory and equipment. Yellow rows are primary keys, light-blue rows are foreign keys, and the grey text shows the data type of each column.

List of All Table Names

#	Table name	What it stores
1	AspNetUsers	Login accounts for all roles (email, password hash, full name, link to customer)
2	Employees	Staff records with employee number, role, retail shop and hire date
3	RetailShops	Retail branches with address, contact number and city
4	Cities	Service cities and the city code used in account numbers
5	LoginAudits	Log of every login attempt (success or failure, IP address, browser)
6	Customers	Customer master data (name, national id, email, contact numbers)
7	CustomerAddresses	Customer addresses with city and primary-address flag
8	CustomerNotifications	Messages sent to the customer panel (bill ready, payment reminder)
9	FeedbackEntries	Customer ratings (1 to 5) and comments with review status
10	ContactMessages	Messages sent from the public Contact page and their reply status
11	Plans	Service plans: type, cycle, speed or hours, charge, deposit, installation
12	Orders	Service orders with number, status, branch, plan and verification details
13	OrderItems	Products or equipment requested in an order with quantity and price
14	OrderStatusHistories	Audit trail of every order status change
15	FeasibilityReports	Technical site-check result for an order
16	Connections	Provisioned customer connection with 16-character account number
17	ConnectionPlanHistories	History of plans assigned to a connection
18	Bills	Customer bills with period, charges, tax, total, paid and due amounts
19	BillItems	Individual line items of a bill
20	Payments	Payments with method, reference, amount and verification status
21	Vendors	Suppliers of equipment and materials
22	Products	Catalogue of equipment and materials with cost and sale price
23	StockItems	Quantity on hand, reserved quantity and reorder level per product
24	StockMovements	Audit trail of every stock in, stock out and adjustment
25	Equipment	Individual equipment units (asset tag, serial, status, assigned connection)
26	EquipmentRequests	Customer cancellation or return requests for equipment

Note: Employee records are matched to their AspNetUsers login by email address, so there is no foreign-key line between these two tables.

Complete ERD (Entity Relationship Diagram)

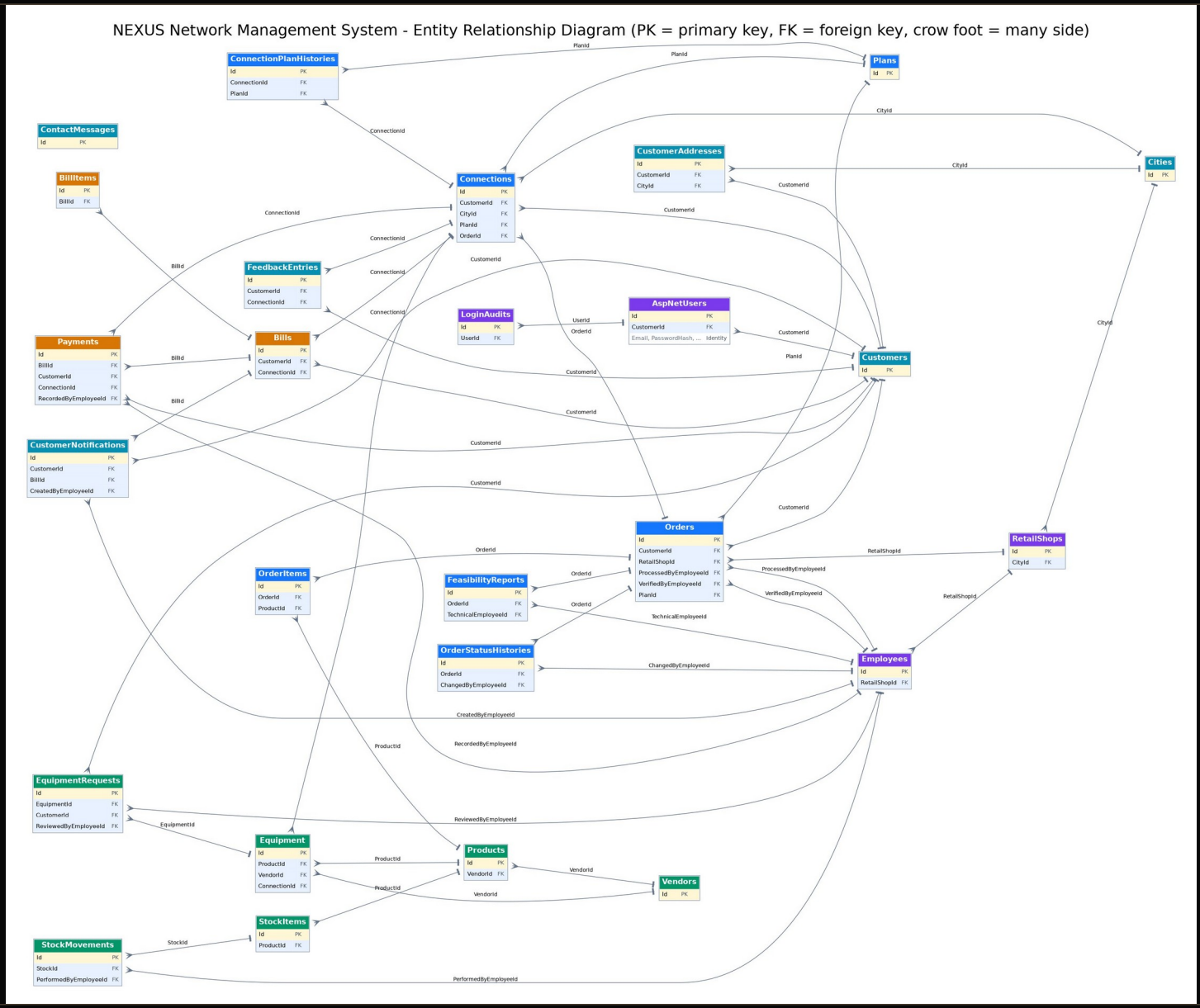


Figure 63: Complete Entity Relationship Diagram of the NEXUS database

Purpose: The ERD shows how the 26 tables are connected. Each line goes from a child table (crow-foot end, many) to its parent table (single end, one) through the foreign key written on the line. For example one Customer has many Orders, Connections, Bills and Payments; one Order has one FeasibilityReport and can create one Connection; one Bill has many BillItems and Payments; and one Product has one Stock record with many StockMovements. Employees appear in many tables because every important action stores the employee who performed it.

Relationship List (Foreign Keys)

#	Child table (many)	Foreign key column	Parent table (one)	Relationship
1	AspNetUsers	CustomerId	Customers	Many-to-One
2	Bills	CustomerId	Customers	Many-to-One
3	Bills	ConnectionId	Connections	Many-to-One
4	BillItems	BillId	Bills	Many-to-One
5	Connections	CustomerId	Customers	Many-to-One
6	Connections	CityId	Cities	Many-to-One
7	Connections	PlanId	Plans	Many-to-One
8	Connections	OrderId	Orders	Many-to-One
9	ConnectionPlanHistories	ConnectionId	Connections	Many-to-One
10	ConnectionPlanHistories	PlanId	Plans	Many-to-One
11	CustomerAddresses	CustomerId	Customers	Many-to-One
12	CustomerAddresses	CityId	Cities	Many-to-One
13	CustomerNotifications	CustomerId	Customers	Many-to-One
14	CustomerNotifications	BillId	Bills	Many-to-One
15	CustomerNotifications	CreatedByEmployeeId	Employees	Many-to-One
16	Employees	RetailShopId	RetailShops	Many-to-One
17	Equipment	ProductId	Products	Many-to-One
18	Equipment	VendorId	Vendors	Many-to-One
19	Equipment	ConnectionId	Connections	Many-to-One
20	EquipmentRequests	EquipmentId	Equipment	Many-to-One
21	EquipmentRequests	CustomerId	Customers	Many-to-One
22	EquipmentRequests	ReviewedByEmployeeId	Employees	Many-to-One
23	FeasibilityReports	OrderId	Orders	Many-to-One
24	FeasibilityReports	TechnicalEmployeeId	Employees	Many-to-One

#	Child table (many)	Foreign key column	Parent table (one)	Relationship
25	FeedbackEntries	CustomerId	Customers	Many-to-One
26	FeedbackEntries	ConnectionId	Connections	Many-to-One
27	LoginAudits	UserId	AspNetUsers	Many-to-One
28	Orders	CustomerId	Customers	Many-to-One
29	Orders	RetailShopId	RetailShops	Many-to-One
30	Orders	ProcessedByEmployeeId	Employees	Many-to-One
31	Orders	VerifiedByEmployeeId	Employees	Many-to-One
32	Orders	PlanId	Plans	Many-to-One
33	OrderItems	OrderId	Orders	Many-to-One
34	OrderItems	ProductId	Products	Many-to-One
35	OrderStatusHistories	OrderId	Orders	Many-to-One
36	OrderStatusHistories	ChangedByEmployeeId	Employees	Many-to-One
37	Payments	BillId	Bills	Many-to-One
38	Payments	CustomerId	Customers	Many-to-One
39	Payments	ConnectionId	Connections	Many-to-One
40	Payments	RecordedByEmployeeId	Employees	Many-to-One
41	Products	VendorId	Vendors	Many-to-One
42	RetailShops	CityId	Cities	Many-to-One
43	StockItems	ProductId	Products	Many-to-One
44	StockMovements	StockId	StockItems	Many-to-One
45	StockMovements	PerformedByEmployeeId	Employees	Many-to-One

Login Credentials

This page lists the login details of every panel of the NEXUS Network Management System. The email address is used as the username, and all accounts share the same password.

Panel Accounts

#	Panel	Role	Email (Username)	Password
1	Admin Panel	Admin	admin@nexos.local	**Nexus@123**
2	Retail Panel	Retail	retail@nexos.local	**Nexus@123**
3	Technical Panel	Technical	technical@nexos.local	**Nexus@123**
4	Accounts Panel	Accounts	accounts@nexos.local	**Nexus@123**
5	Customer Panel	Customer	jhon@gmail.com	**Nexus@123**

> Note: Each user can open only his own panel after login. The Customer account (jhon@gmail.com) opens the Customer Panel; the other four accounts open the Admin, Retail, Technical and Accounts panels.